

Audience Effect in Microfinance:

Does Social Observability of Default Increase Repayment?

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Abstract

Does making default observable to peers increase repayment in microfinance? Many microfinance institutions have moved from joint to individual liability contracts, yet continue to collect installments in group settings that preserve the *social observability of default* as an institutional feature. In a framed lab-in-the-field experiment, I vary whether default is observable to peers along with the borrower's realized income, for a hypothetical loan on which borrowers make incentivized repayment decisions. The experiment was conducted with 919 agricultural borrowers across 100 borrowing groups in Bangladesh, and the within-subject variation in observability identifies the *audience effect* on repayment. I find that an audience raises repayment by 3.3 to 5.3 percent relative to the amount repaid without an audience. While the audience effect does not vary detectably across income states, repayment without an audience rises with income, ranging from half to two-thirds of the debt obligation. Experimental choices are consistent within borrowers, with reliability coefficients between 0.81 and 0.87, indicating systematic rather than random responses. Since the experiment removes other contractual penalties, the conceptual framework interprets repayment without an audience as reflecting intrinsic motivation and the additional repayment induced by an audience as reflecting reputational motivation. Although the estimated repayment gain is modest relative to the total debt obligation, a reduction in field default of the same magnitude would raise lender profit by 34 to 55 percent of the zero-default margin.

Keywords: microfinance, individual liability, lab-in-the-field experiment, social observability, audience effect, behavioral motivation, reputation

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1 Introduction

Microfinance expanded access to formal credit by substituting physical collateral with social and relational enforcement mechanisms. Early joint liability contracts relied on collective financial responsibility to induce peer monitoring, social sanction and assortative matching (Stiglitz, 1990; Besley and Coate, 1995; Ghatak, 2001). Many lenders subsequently shifted toward individual liability, partly to reduce the coordination burdens and shared default risk associated with joint contracts (Madajewicz, 2011; Giné and Karlan, 2014). While this shift removed shared repayment obligation, it did not eliminate the group structure of lending. In institutions such as ASA and Grameen Bank II, borrowers from the same village neighborhoods form borrowing groups, and group based repayment collection and follow up practices expose delinquency to peers and the wider community (Armendáriz and Morduch, 2010). The persistence of these practices in individual liability contract suggests that lenders may expect the social observability of default to strengthen repayment incentives.

Does making default socially observable increase repayment under individual liability microfinance? Whether it does so, remains difficult to establish because observability is often a fixed feature of such lending environment and provides little empirical variation. Moreover, individual liability contracts retain pecuniary incentives such as termination threats and progressive lending to sustain repayment.¹ Unlike these pecuniary incentives, social observability does not directly alter the monetary payoff from default, but makes nonpecuniary motivations salient. Akerlof (1980) shows that deviation from social norms generates disutility through community disapproval and, for some individuals, through the violation of internalized values. Andreoni and Bernheim (2009) similarly show that individuals may care not only about behaving fairly but also about being perceived as fair, which they describe as an *audience effect*. Adapted to the microfinance context, these mechanisms imply that repayment may be sustained by a self-imposed obligation to honor the loan, while lender-imposed social observability of default may add concerns about reputational loss or being perceived as violating established norms. The experiment distinguishes repayment when default is not socially observable from the additional repayment induced by observability, which I interpret in the conceptual framework as reflecting intrinsic and reputational motivations, respectively.

To study these motivations, I conduct a framed lab-in-the-field experiment with 919 active BRAC microfinance borrowers across 100 villages in southern Bangladesh. Borrowers face a realistic rice farming scenario in which they choose how much of a fixed hypothetical loan to repay. Greater default leaves more experimental income, which is converted into cash payments to incentivize repayment decisions. The key experimental variation is whether default is socially observable. In the private

¹Termination incentives and progressive lending have been studied theoretically and empirically by Armendáriz and Morduch (2000), Giné et al. (2010b), and Giné et al. (2010a). Frequent repayment schedules have also been viewed as a means of cultivating financial discipline, although Field and Pande (2008); McIntosh (2008) find little difference in default rates when installment frequency varies. This paper focuses on behavioral motivations in the absence of these pecuniary incentives.

game (P), default remains confidential, providing a benchmark measure of repayment associated with intrinsic motivation. In the audience game (A), borrowers are informed that their default amount will be revealed to members of their group, thereby introducing an audience for repayment behavior and allowing me to identify the effect of social observability on repayment. Sessions were conducted at the respective village meeting centers with groups of 6 to 12 borrowers from actual microfinance groups, closely resembling the setting in which real loan transactions occur. The experiment also varies realized experimental income across good, normal, and bad states, holding the debt obligation fixed. The order of the private and audience games is randomized at the session level.

I find that an audience increases repayment by 3.3 to 5.3 percent relative to repayment without an audience. The OLS fixed effects estimates show that audience raises observed repayment by 3.3–3.5 percent across income states. The estimates on observed repayment may understate the underlying latent effect, because repayment is censored at the lower or upper limit for up to two-fifths of observations. I therefore complement them with a two-limit Tobit specification that accounts for censoring and identifies a latent audience effect of 3.8–5.3 percent. A Monte Carlo exercise calibrated to the experimental design shows that censoring attenuates the observed effect relative to the latent effect, while the Tobit estimator recovers the latent effect with little bias. The estimates further show that repayment is state dependent, increasing with realized income. Without an audience, borrowers repay roughly one-half to two-thirds of the debt obligation, with repayment declining in the adverse state even though full repayment remains feasible. The audience effect itself does not vary detectably across income states, indicating that it adds to a state-dependent benchmark rather than substituting for it.

Since each borrower makes repeated decisions in a single session, two concerns arise. The first is whether repeated choices reflect systematic borrower behavior rather than idiosyncratic responses. Repayment choices are highly consistent within borrowers across states and rounds, with reliability coefficients ranging from 0.81 to 0.87, well above those generated under random choice. The second is whether the audience effect changes over the course of the session or depends on the order in which the conditions are presented. Repayment levels decline across rounds, but the audience effect does not vary across rounds, and the randomized game order reveals no detectable order asymmetry. Round fixed effects absorb the common decline in repayment across rounds, while the audience effect is identified from the within-borrower difference between the private and audience conditions. I therefore report the pooled coefficient as the average audience effect across rounds.

Although modest relative to the total debt obligation, the repayment gain can be economically important for lender profitability (Cull et al., 2009; Rosenberg et al., 2013; Flatnes and Carter, 2019). When lending margins are thin, a small reduction in default can generate a large proportional increase in profit. The estimated increase in observed expected repayment of 675.5 Taka corresponds to a 1.93 percentage-point reduction in the unpaid share of the total debt obligation. If a default reduction of

the same magnitude occurred on an otherwise comparable field loan, it would generate a profit gain equivalent to 34 to 55 percent of the zero-default profit margin.²

The paper makes three contributions. First, it contributes to the microfinance literature by examining socially embedded repayment incentives that persist in individual liability contracts. The large literature on microfinance enforcement has focused primarily on joint liability contracts, where shared financial responsibility creates incentives for peer monitoring, social sanctions and assortative matching (Stiglitz, 1990; Besley and Coate, 1995; Ghatak, 2001; Armendáriz and Morduch, 2000; Giné et al., 2010a; Balboni et al., 2022). Yet individual liability lending often retains group-based practices that make default observable to peers even without any financial ties. I isolate the effect of this remaining social feature and show that making default observable increases repayment. The closest related study is Dhimi et al. (2022), who show that shame increases ex-ante effort in individual liability contracts. I complement their evidence by identifying the effect of observability on ex-post repayment among borrowers in actual microfinance groups.

Second, the paper brings behavioral theories of norm driven decision making to the study of repayment incentives in a formal credit market. A large experimental literature uses social preference games, including dictator, ultimatum, public goods, and trust games, to show how norms, fairness, and social image concerns influence decision making (Kimbrough and Vostroknutov, 2016; Dana et al., 2007; Andreoni and Bernheim, 2009). This paper extends these behavioral insights to an incentivized repayment decision framed around a formal credit relationship. The experiment separates repayment when default remains private from the additional response induced when default becomes socially observable. Private repayment varies with the borrower's realized income, while the audience effect does not. This distinction suggests that the benchmark repayment decision is conditional on economic circumstances, while social observability provides an additional reputational incentive across those circumstances. This is consistent with Dana et al. (2007), who show that norm conforming behavior weakens when "moral wiggle room" is available. The results therefore connect theories of intrinsic and reputational motivation to a contract relevant economic decision.

Third, the paper provides a within-subject design for recovering individual specific measures of repayment behavior. Repeated repayment decisions under private and social observability conditions allow these responses to be characterized at the borrower level rather than only in the aggregate. In the conceptual framework, I interpret these measures as reflecting intrinsic and reputational motivations as reduced-form behavioral measures from the experiment. These borrower level measures provide a basis for future work linking behavioral repayment motives to field outcomes such as borrowing and demand for risk management products.

²The accounting exercise does not account for additional costs of implementing social observability. Such costs may be limited where lenders already collect repayments through village meetings or household visits, but observability could still require additional staff time or affect borrower retention and group relationships.

The rest of the paper proceeds as follows. Section 2 presents the conceptual framework, Section 3 describes the experimental design, and Section 4 outlines the estimands and empirical strategy. Section 5 reports the main results, Section 6 examines heterogeneity, Section 7 discusses their implications for lender profitability, and Section 8 concludes with interpretation, and policy implications.

2 Conceptual Framework

The conceptual framework adapts the characterization of behavioral motivation in Bénabou and Tirole (2006) to the institutional setting of individual liability microfinance. I model repayment as a tradeoff between the monetary utility from nonrepayment and the disutility from deviation from intrinsically and reputationally motivated norms.

2.1 A Model of Behavioral Motivations for Repayment

Let borrower i choose repayment amount $r \in [0, \bar{r}]$ after observing her income $y(\theta_i)$, where $\bar{r}$ is the fixed debt obligation and $\theta_i \in [\underline{\theta}, \bar{\theta}]$ indexes the income realization, with $y'(\theta_i) > 0$. Let $s \in \{0, 1\}$ denote whether the borrower's default is socially observable, or equivalently, whether there is an audience for r . The borrower's utility consists of three components,

$$U(r; s, \theta_i, \mathbf{x}_i) = \underbrace{u(y(\theta_i) - r)}_{\text{utility from retained earning}} + \underbrace{\alpha_i N(r - n^I(\theta_i))}_{\text{disutility from violating the intrinsic norm}} + s \underbrace{\gamma_i N(r - n^R(\mathbf{x}_i))}_{\text{disutility from violating the reputational norm}} \quad (1)$$

with $u'(\cdot) > 0$ and $u''(\cdot) < 0$. The disutility from violating the intrinsically motivated norm depends on $n^I(\theta_i)$, the borrower's intrinsic belief about how much she ought to repay in state θ_i , and on $\alpha_i \geq 0$, the value she places on adhering to her intrinsic value. This intrinsic motivation affects repayment regardless of whether default is observed by others. The disutility from violating the reputationally motivated norm depends on $n^R(\mathbf{x}_i)$, what the borrower believes others expect her to repay given her observable characteristics $\mathbf{x}_i$, and on γ_i , the value she places on adhering to the reputational expectation. This reputational motivation enters only when $s = 1$, when default is socially observable. The penalty function is common to both norms, $N(z) = \tilde{N}(\min\{z, 0\})$, with $N'(z) \geq 0$ and $N''(z) \leq 0$ for $z < 0$. Only repayment below the relevant norm generates disutility, while repayment at or above the norm generates no additional utility. The utility function collapses to the two conditions,

$$(s = 0) : \quad U = u(y(\theta_i) - r) + \alpha_i N(r - n^I(\theta_i))$$

$$(s = 1) : \quad U = u(y(\theta_i) - r) + \alpha_i N(r - n^I(\theta_i)) + \gamma_i N(r - n^R(\mathbf{x}_i)).$$

Without social observability, optimal repayment $r^*(0)$ reflects the trade off between the monetary gain

from retaining income and the disutility from repaying below the intrinsically motivated norm. With social observability, optimal repayment $r^*(1)$ reflects the same trade off and the additional disutility from repaying below the reputationally motivated norm. The repayment gain from social observability is $\Delta_i \equiv r^*(1) - r^*(0)$, which I refer to as the *audience effect* in the reduced form analysis.

The conceptual framework does not explicitly model posterior inference by peers. In a richer signaling framework, peers would observe repayment, form a judgment about the borrower, and the borrower would anticipate that judgment when choosing how much to repay. I instead summarize the repayment norm she believes others expect her to meet through $n^R(\mathbf{x}_i)$ and the value she places on adhering to that norm through γ_i . Borrowers may similarly differ in the intrinsic norm $n^I(\theta_i)$ and in the value they place on adhering to it, α_i . Thus, intrinsic and reputational motivations may differ across borrowers because of differences in the relevant repayment norm, the value placed on adhering to it, or both. The reduced-form analysis therefore estimates the composite effect of these motivations on repayment rather than separately identifying their underlying components.

2.2 Comparative Statics

Social Observability. Assume $u'(\cdot) > 0$, $u''(\cdot) < 0$, $N'(\cdot) > 0$, and $N''(\cdot) \leq 0$ over the relevant range, so that utility is concave in repayment and the second-order condition $u''(\cdot) + \alpha_i N''(\cdot) < 0$ holds. The first-order conditions for an interior optimum are,

$$-u'(y(\theta_i) - r^*(0)) + \alpha_i N'(r^*(0) - n^I(\theta_i)) = 0 \quad (2)$$

$$-u'(y(\theta_i) - r^*(1)) + \alpha_i N'(r^*(1) - n^I(\theta_i)) + \gamma_i N'(r^*(1) - n^R(\mathbf{x}_i)) = 0 \quad (3)$$

In both cases, the borrower equates the marginal monetary cost of repayment with the marginal disutility from norm violation. With social observability, an additional marginal cost arises from violating the reputational repayment norm through $\gamma_i N'(r^*(1) - n^R(\mathbf{x}_i))$.

Proposition 1 (Effect of social observability on repayment). *Let $r^*(0)$ and $r^*(1)$ denote the optimal repayment levels without and with social observability, respectively. If $\gamma_i > 0$ and $N'(r^*(1) - n^R(\mathbf{x}_i)) > 0$ at an interior optimum, then introducing an audience strictly increases optimal repayment, so that $r^*(1) > r^*(0)$. Equality holds only at a corner solution or where $N'(\cdot) = 0$. The individual repayment gain $\Delta_i \equiv r^*(1) - r^*(0)$ is weakly increasing in γ_i , and the average audience effect is $\mathbb{E}[\Delta_i]$.*

Income State. Applying the implicit function theorem to equations (2) and (3) gives,

$$\frac{\partial r^*(0)}{\partial \theta_i} = \frac{u''(y(\theta_i) - r^*(0))y'(\theta_i) + \alpha_i N''(r^*(0) - n^I(\theta_i)) \frac{dn^I(\theta_i)}{d\theta_i}}{u''(y(\theta_i) - r^*(0)) + \alpha_i N''(r^*(0) - n^I(\theta_i))}. \quad (4)$$

$$\frac{\partial r^*(1)}{\partial \theta_i} = \frac{u''(y(\theta_i) - r^*(1))y'(\theta_i) + \alpha_i N''(r^*(1) - n^I(\theta_i)) \frac{dn^I(\theta_i)}{d\theta_i}}{u''(y(\theta_i) - r^*(1)) + \alpha_i N''(r^*(1) - n^I(\theta_i)) + \gamma_i N''(r^*(1) - n^R(\mathbf{x}_i))}. \quad (5)$$

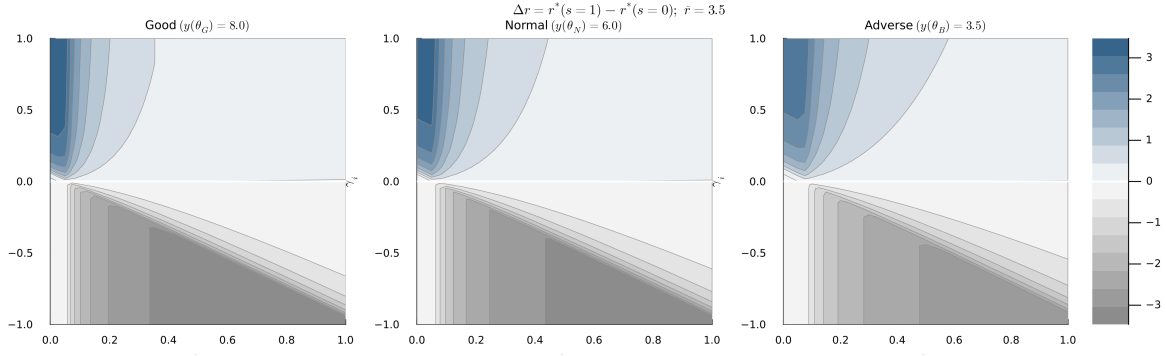
Without socially observable default in equation (4), the denominator is negative under $u'' < 0$ and $N'' \leq 0$. The numerator contains two components. The first, $u''(y(\theta_i) - r^*(1))y'(\theta_i) < 0$ when $y'(\theta_i) > 0$, reflects an income effect. A better income state raises available resources, lowers the marginal utility of retained income, and therefore lowers the opportunity cost of repayment. The second term is non-positive when $dn^I(\theta_i)/d\theta_i \geq 0$, reflecting an intrinsic norm response. If the borrower's intrinsic repayment norm rises with the income state, deviating from that norm becomes more costly, further increasing the marginal utility of repayment. Thus, the numerator is non-positive and $\partial r^*(0)/\partial \theta_i \geq 0$.

The same two components determine state dependence when default is socially observable. The denominator of equation (5) remains negative and additionally reflects the curvature of the disutility from violating the reputational norm. Because the reputational repayment norm $n^R(\mathbf{x}_i)$ depends on observable characteristics rather than the income state, $\partial n^R(\mathbf{x}_i)/\partial \theta_i = 0$. The reputational norm therefore affects the magnitude of the response through the denominator but creates no separate state-dependence in the numerator. Hence, $\partial r^*(1)/\partial \theta_i \geq 0$ under the same conditions and $r^*(1) \geq r^*(0)$ when $\gamma_i > 0$.

Proposition 2 (State dependence of repayment). *Suppose $u'' < 0$, $N'' \leq 0$, $y'(\theta_i) > 0$, and $dn^I(\theta_i)/d\theta_i \geq 0$. Then optimal repayment is weakly increasing in θ_i both without and with social observability, so that $\partial r^*(0)/\partial \theta_i \geq 0$ and $\partial r^*(1)/\partial \theta_i \geq 0$. Borrowers repay less in worse income states because adverse states raise the marginal value of retained income against a lower cost of violating the intrinsic repayment norm. Since $n^R(\mathbf{x}_i)$ does not vary with the income state, social observability introduces no separate state-dependent repayment gain.*

Repayment gain in α_i and γ_i space. Although the main model assumes $\gamma_i \geq 0$, the illustration allows $\gamma_i < 0$ to represent the scenario where social observability makes lower repayment reputationally valuable. For example, lower repayment could signal bargaining strength, financial prudence, prioritization of household needs, or conformity with a local norm that is less demanding than the contractual obligation. Appendix B.1 uses explicit functional forms to illustrate how the repayment gain varies over the (α_i, γ_i) parameter space.

Figure 2.1: Repayment gain in α_i and γ_i space



Heterogeneity in repayment. The comparative statics examine the effect of social observability holding socioeconomic characteristics fixed. Borrower characteristics may nevertheless be associated with both private repayment and the audience effect through the behavioral parameters of the model. Borrowers may differ in α_i , the value they place on adhering to their intrinsic norm, which may be shaped by upbringing, socialization, and past experience. Education may also be associated with intrinsic and reputational motivations. More educated borrowers may place less value on adhering to repayment as an unconditional internal obligation, while placing greater value on adhering to the reputational norm when default is observable.

Considering γ_i , borrowers who place greater value on adhering to the reputational norm have a stronger incentive to avoid observable default, but economic circumstances may affect this value in competing ways. Those who expect to rely on informal support may place greater value on maintaining a favorable reputation, while better-off borrowers may place greater value on signaling good standing within the community. Exposure to a common shock may have a similar effect. It may make the need for community support more salient and therefore strengthen reputational motivation, or provide a socially accepted justification for lower repayment that weakens it.

Borrower characteristics may also shift $n^R(\mathbf{x}_i)$, the repayment norm that borrowers believe others expect them to meet. If community members expect those with greater repayment capacity to meet a higher norm, observability may create a larger response among such borrowers even when they place no greater value on adhering to that norm.

These mechanisms may offset one another, and so the empirical analysis identifies their combined effect through Δ_i rather than separately identifying variation in $n^R(\mathbf{x}_i)$ from variation in γ_i . I interpret the heterogeneity results as reduced-form evidence on how the effect of social observability varies across borrowers.

3 Experimental Procedures and Data

This section maps the behavioral motivations in the conceptual framework to the experimental conditions and describes the experimental procedures, data collection, and sample.

3.1 Institutional Setting and Mapping to Experiment

BRAC, one of the world’s largest microfinance providers, operates individual liability lending in Bangladesh. Although borrowers are individually responsible for their loans, borrowing groups are formed in the village neighborhood. Borrowers are required to meet periodically at village centers, where loan officers collect installments, or loan officers visit borrowers at their homes. These practices, in a close knit community make repayment and delinquency visible to peers and the broader community, inducing reputational motivation to operate.

Table 3.1 summarizes the mapping between the behavioral motivations in the conceptual framework and the experimental design. Since the experiment removes other contractual penalties, repayment when default remains private provides a measure of repayment associated with intrinsic motivation. The additional repayment induced when default becomes observable to peers identifies the audience effect, which I interpret as a reputationally motivated response. These repayments represent the conceptual framework’s interpretation of the reduced-form responses rather than separately identified behavioral mechanisms.

Table 3.1: Mapping behavioral motivations to microfinance context and experimental conditions.

Behavioral Motivation ^a	Microfinance Context	Experimental Condition
<i>Intrinsic</i> : Motivation to undertake an action because it is valued in itself or is consistent with an internalized self-standard	Borrower’s internalized sense of obligation to honor a loan	<i>Private Game (P)</i> : Borrowers privately decide how much to repay and are reminded only of the debt obligation.
<i>Reputational</i> : Motivation to undertake an action because of how it would be perceived or evaluated by others	Borrower’s motivation to maintain her social standing or reputation	<i>Audience Game (A)</i> : Borrowers are informed that their default will be revealed to peers.

^a The definitions draw on Andreoni and Bernheim (2009), Akerlof (1980), Bénabou and Tirole (2006), Cialdini et al. (1991), and Cialdini and Goldstein (2004). The experiment varies the social observability of default.

3.2 Experimental Design

The experiment presents a rice farming scenario in which each game represents a farming season and borrowers make repayment decisions under good, normal, and bad income realizations (Figure A.4). The hypothetical loan obligation is calibrated to borrowing for farming in the study population, while

monthly installments are collapsed into a single end-of-season repayment decision to simplify the experimental task. Decisions are incentivized with real monetary payments. Sessions were conducted at the village meeting centers with borrowers from belonging to same actual microfinance groups, resembling the setting in which real loan transactions occur (Figure A.3).

In each game, borrowers face a fixed debt obligation of 35,000 Taka and choose repayment in increments of 5,000 Taka between zero and the full obligation. Income is 80,000 Taka in the good state, 60,000 Taka in the normal state, and 35,000 Taka in the bad state, with corresponding probabilities of 0.30, 0.40, and 0.30. Borrowers therefore have sufficient income to repay the full obligation in all three states, although the marginal cost of repayment varies with income.

In the private games, repayment and default remain confidential. Participants are reminded of their debt obligation and that they may retain the proportional monetary gains from defaulting. In the audience games, participants are informed that their default amount will be revealed to members of their borrowing group, but not the income state in which that default amount was chosen. Peers therefore would observe the default but not the underlying income state. To make social observability salient, a board listing all participants' names is displayed with space to record individual default amounts (Figure A.5). Participants are informed that one audience game decision will be drawn at random for revelation after all repayment decisions are complete. In practice, no default amounts were revealed at the end of the session. Participants therefore made their decisions under the expectation of revelation, while actual disclosure was withheld to avoid extending any reputational consequences beyond the experimental setting.

I use a strategy method in which each borrower makes repayment choices for all combinations of game type, round, and income state before payoff determination. Each participant therefore makes repayment decisions across $2 \times 4 \times 3 = 24$ hypothetical scenarios, including the practice round. To mitigate anchoring and order effects, the sequence in which participants were presented with the private and audience games is randomized at the session level. The order of income states and the contextual examples read by facilitators were also randomized across sessions.

Table 3.2: Game sequence across rounds

	Practice		Actual Game						
	Round 1		Round 2		Round 3		Round 4		Payoff Determination
	G1	G2	G3	G4	G5	G6	G7	G8	
Sequence 1	P	A	A	P	P	A	A	P	Draw round-state pair for each game condition
Sequence 2	A	P	P	A	A	P	P	A	

After all repayment decisions are completed in the session, one state-round pair is randomly drawn from each game condition to determine experimental earnings. Income remaining after repayment in

the selected private and audience decisions is converted at a rate of 1/100. Participants also receive a 150 Taka ensuring a positive payment regardless of the selected repayment decisions. The conversion rate was chosen so that expected experimental earnings were approximately equivalent to a day's agricultural wage.

The design abstracts from dynamic incentives in the credit contract, that is, repayment in one round does not affect future borrowing opportunities, and the game facilitators ensured that no loan officers were present during the sessions. Thus, differences in repayment across game types should not be attributed to concerns about future loan access or the presence of authoritative figures. By varying social observability within subjects while holding constant loan size, income risk, and monetary incentives, the experiment isolates the audience effect on repayment behavior.

3.3 Sampling and Data Collection

The experiment was conducted between January and April 2025 with active BRAC microfinance borrowers in southern Bangladesh. Data collection was organized and implemented by the BRAC Institute of Governance and Development (BIGD) in collaboration with BRAC and the University of California, Davis. Participants were drawn from a broader randomized trial evaluating indexed risk-management tools among agricultural microfinance borrowers. The broader RCT includes 6,683 borrowers from 255 BRAC branches across 13 districts (Figure A.1). Branches were randomly selected within districts, and village organizations (VOs) were randomly selected within agriculturally intensive areas. Within selected VOs, a screening questionnaire was administered to BRAC microfinance borrowers to select farm households eligible for the broader RCT. BRAC offers small individual liability microfinance loans only to female members, so both the broader RCT population and the experimental sample drawn from it consist entirely of women.

For the repayment experiment, I randomly selected 120 branches from the broader RCT, stratified by its three treatment arms, and randomly selected one VO within each branch. All agricultural borrowers in the broader RCT sample belonging to the selected VO were invited to participate in the experimental session. Approximately [X] borrowers were invited to the sessions included in the final experimental design, of whom 919 attended and participated, corresponding to an attendance rate of [X] percent. Participation was voluntary, and no additional sample restrictions were imposed based on borrower characteristics or repayment behavior. The experimental sample therefore consists of participating BRAC agricultural borrowers drawn from the broader RCT population in the study area.

The study initially targeted 1,200 borrowers across 120 VOs. The first 20 sessions used an earlier version of the design with a single round, and a fixed ordering of the Private and Audience games. I exclude these sessions because they do not share the repeated round and randomized game sequence structure of the final design. The analysis sample therefore consists of 919 borrowers from the re-

maintaining 100 VOs, all of whom participated in the four-round design described in Section 3.2. Prior to field implementation, 13 pilot and facilitator-training sessions were conducted with members of the research team present to observe implementation and provide structured feedback. Across the pilot, initial, and final designs, 133 sessions were conducted at 120 village locations. Figure A.2 summarizes the resulting experimental sample.

Sessions were administered by nine trained BIGD field supervisors with prior experience conducting household surveys and coordinating enumerator teams in the study region. All held at least a bachelor's degree and received training in the experimental procedures before field implementation. The study protocol was approved by the Institutional Review Board of the BRAC James P Grant School of Public Health, BRAC University (Protocol IRB-2024-ES-37), and all participants provided oral informed consent before the experiment.

Table 3.3 reports sample characteristics, including borrower demographics, farming activities, credit access, recent shock exposure, and experience with crop insurance. The characteristics are drawn from the baseline survey of the broader RCT. Each session included 6 to 10 borrowers, with an average of approximately nine participants, drawn from an existing VO.

Table 3.3: Sample Description

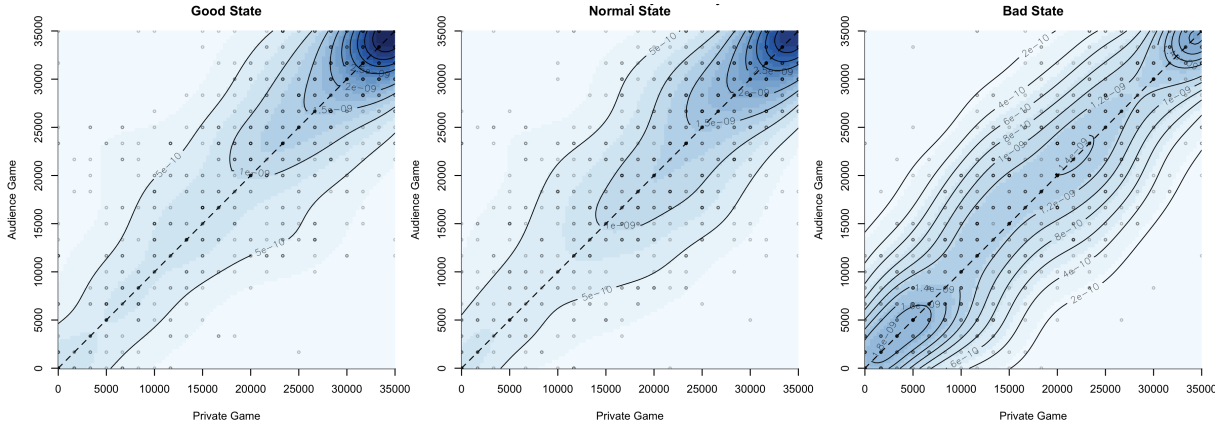
	% or Mean	S.D.	N
<i>Respondent characteristics</i>			
Age in years	40.38	10.07	919
Participates in earning activity (%)	71.38		905
Grade 9 or higher education (%)	15.56		919
Food insecurity score	9.16	6.21	919
Risk preference	5.83	4.39	919
<i>Farming type</i>			
Rice farmer, Aman (%)	71.93		919
Non-rice farmer, Aman (%)	50.27		919
<i>Credit</i>			
Total BRAC loan (Taka)	67,310	60,434	919
Total loan (Taka)	199,098	208,558	919
Risk rationed (%)	21.87		919
Quantity rationed (%)	35.47		919
<i>Shock exposure</i>			
Flood, last season (%)	43.42		919
Pest, last season (%)	17.63		919
Household dispersion within VO (km)	0.254	0.234	919
<i>Insurance & future interest (Boro)</i>			
Purchased crop insurance (%)	13.33		285
Interested in buying crop insurance (%)	79.95		848
<i>Borrowing group information</i>			
Participants in the experiment	9.19	.97	100
Household dispersion within VO (km)	0.254	0.234	100

Notes: Binary variables are reported as percentages and continuous variables as means with standard deviations. Missing values in these descriptive characteristics do not affect the sample used in the main experimental repayment analysis.

3.4 Descriptive Statistics

Figure 3.1 presents the joint distribution of mean private game repayment and mean audience game repayment, averaged across game rounds and Table A.1 provides the distribution of repayment by round, audience, and income states. The joint density is concentrated slightly above the 45-degree line, reflecting higher repayment under the audience game. In the favorable and normal yield states, mass concentrates near the full-repayment boundary, reflecting a large share of borrowers repaying the full loan amount under both conditions. In the bad yield state, this concentration dissipates and the density exhibits two distinct modes near the lower and upper bounds of the repayment choice set, consistent with a bimodal distribution of repayment behavior under adverse income realizations. The joint density further reveals substantial heterogeneity across borrowers in both the level of private game repayment and the magnitude of the audience effect, which motivates recovering individual-level norm measures.

Figure 3.1: Joint distribution of mean private game and audience game repayments.



4 Empirical Strategy

I exploit the within-subject design in which each borrower i makes repayment decisions across rounds r and game conditions $g \in \{P, A\}$. Let y_{igr}^k denote repayment under income state $k \in \{G, N, B\}$, corresponding to the good, normal, and bad states. The associated state probabilities are $p^G = 0.30$, $p^N = 0.40$, and $p^B = 0.30$. Expected repayment is defined as the probability-weighted repayment across states, $y_{igr}^E = \sum_{k \in \{G, N, B\}} p^k y_{igr}^k$. For simplicity, let y_{igr} denote the repayment outcome corresponding to a given state. I exclude the first round, introduced as a practice round. The analysis sample contains 18 observations per borrower, corresponding to two game conditions, three analysis rounds, and three income states.

4.1 Fixed effects estimation of observed repayment

Identification comes from within-borrower variation between the private and audience conditions. I estimate the following fixed effects specification,

$$y_{igr} = \beta \text{Audience}_g + \mu_i + \lambda_r + \varepsilon_{igr}, \quad (6)$$

where y_{igr} denotes the repayment amount chosen by borrower i in game g during round r , and Audience_g equals one for the Audience game and zero for the Private game. The coefficient β captures the within-borrower difference in observed repayment between the audience and private games, which I refer to as the audience effect as the response to social observability of default. Individual fixed effects μ_i absorb all time-invariant heterogeneity across borrowers, including baseline behavioral motivations, demographic characteristics, and session specific factors. Round fixed effects λ_r account for common changes over the course of the experiment, such as learning or fatigue. Standard errors are clustered at the session level to allow for heteroskedasticity and correlation across repeated decisions by the same borrower and across borrowers within the same experimental session. Session-level dependence is plausible because borrowers within a session share the experimental environment, facilitator, and game sequence assignment.

4.2 Two-limit Tobit estimation of latent repayment

The distribution of repayment choices shown in Figure A.6 exhibits mass at both boundaries of the choice set, with approximately 12 percent of observations at the lower limit of zero and 35 percent at the upper limit of $\bar{y} = 35,000$ Taka. Since observed repayment is bounded at both limits, the linear fixed effects specification may underestimate the underlying behavioral response when latent repayment lies outside the feasible choice set. To account for this censoring, I estimate a two-limit Tobit specification for latent repayment y_{igr}^* . Observed repayment satisfies $y_{igr} = \max\{0, \min\{\bar{y}, y_{igr}^*\}\}$.

The latent repayment equation and associated censoring rule are

$$y_{igr}^* = \beta^T \text{Audience}_g + \mathbf{x}_{igr}' \boldsymbol{\psi} + \varepsilon_{igr}^*,$$

$$y_{igr} = \begin{cases} 0 & \text{if } y_{igr}^* \leq 0, \\ y_{igr}^* & \text{if } 0 < y_{igr}^* < \bar{y}, \\ \bar{y} & \text{if } y_{igr}^* \geq \bar{y}. \end{cases} \quad (7)$$

where $\mathbf{x}_{igr}$ is a vector of controls including round and game sequence indicators. I estimate the two-limit Tobit model by maximum likelihood method, assuming a normally distributed latent error term.

Standard errors are clustered at the session level, as in the fixed effects specification. Individual fixed effects are not included because the incidental parameters problem renders the fixed effects Tobit estimator inconsistent in short panels. The coefficient β^T estimates the audience effect on latent repayment after accounting for censoring at both bounds.

The fixed effects and Tobit specifications therefore estimate observed and latent effects respectively and rely on different assumptions. Appendix E evaluates the finite sample performance of both estimators in Monte Carlo simulations calibrated to the experimental design. The simulations compare the observed fixed effects estimate with the latent audience effect, while varying borrower heterogeneity and dependence across repeated decisions.

4.3 Within-Subject Design Justification

The within-subject design uses repeated repayment choices across rounds and experimental conditions. I first assess whether borrowers' relative repayment behavior is consistent across repeated decisions. I then examine whether the audience effect varies across rounds or depends on game order. Appendix C.1 shows substantial consistency in repayment choices within both the private and audience games. Standardized Cronbach's alpha ranges from 0.791 to 0.809 across income states and from 0.810 to 0.869 across rounds, with all coefficients well above those generated under random choice. Appendix C.2 then examines changes in repayment over rounds, variation in the audience effect across rounds, and differences by game order.

Temporal stability. A concern in a within-subject design is that repayment may change with position in the session. Appendix C.2 shows that repayment declines from round 2 to round 4 in every income state and under both experimental conditions, with round effects jointly different from zero in all three states. List (2025) notes that common time-varying factors can be accounted for using time fixed effects when they do not systematically alter the treatment effect. The main specifications therefore include round fixed effects. Appendix C.2 also shows that the round-by-audience interactions are jointly indistinguishable from zero in all three states, although they are imprecisely estimated. All decisions are made within a single session, and round order cannot be randomized. The round effects therefore cannot be separated from fatigue, learning, or other changes over the session.

Causal transience. A further concern is that behavior under the current experimental condition may be affected by exposure to the other condition through anticipation or carryover (List, 2025). The design limits one important source of carryover by delaying disclosure until all repayment decisions are complete. Borrowers therefore do not observe peer reactions to disclosed default before making subsequent choices. This removes the peer-feedback channel through which an audience decision could directly affect later choices. The order of the private and audience games is randomized at the session level, which with two conditions corresponds to the two-period Latin square discussed by List

(2025) and provides a diagnostic for order effects. Appendix C.2 shows no detectable difference in the audience effect across orders, although the estimates are not sufficiently precise to rule out moderate order effects. Anticipation remains a concern that cannot be tested with these data.

5 Results

This section presents the main repayment results. I first estimate the audience effect on observed repayment using the fixed effects specification, and then estimate the latent audience effect accounting for censoring at the repayment bounds using the two-limit Tobit specification. I subsequently examine whether repayment and the audience effect vary across income states.

5.1 Fixed Effects Estimates

Table 5.1 reports fixed effects estimates of the effect of social observability on repayment separately for each realized income states. Audience increases observed repayment by 562 to 798 Taka across income states. In the conceptual framework, these estimates provide reduced form evidence of the average audience effect, $\mathbb{E}[\Delta_i]$ relecting reputational motivation. The estimated effect is smallest in the bad state, at 562 Taka, and larger in the good and normal yield states, at 767 and 798 Taka respectively. Relative to repayment without an audience, these effects correspond to approximately 3.3 to 3.5 percent. The audience effect is positive and statistically significant in all states. Since monetary incentives are held fixed across the private and audience games, the positive audience effect reflects participants giving up additional earnings in response to the social observability of default. Appendix Table D.1 reports estimates restricted to rounds 2 and 3 and specifications using individual-level clustering, both of which are consistent with the main results.

Repayment in the private game varies substantially across states of nature. Average repayment is lowest in the bad yield state and higher in the normal and good yield states. In the conceptual framework, these estimates reflect changes in borrowers' intrinsic motivation to repay as the income state changes. A lower income realization increases the marginal utility of retained income, so repayment becomes more costly at the margin. It may also lower the borrower's intrinsic belief about what she feels obligated to repay given competing needs under adverse circumstances. Whether private repayment and the audience effect vary across income states is formally tested in Section 5.3.

Table 5.1: Audience Effect on Repayment: Fixed effects Estimates

	(1) <i>Good Yield Repayment</i>	(2) <i>Normal Yield Repayment</i>	(3) <i>Bad Yield Repayment</i>	(4) <i>Expected Repayment</i>
Audience	767.14*** (197.70)	797.97*** (248.96)	562.21*** (198.45)	717.99*** (147.81)
Constant	23,108.45*** (98.85)	22,629.67*** (124.48)	17,261.52*** (99.22)	21,162.86*** (73.91)
Audience effect / private mean	3.32%	3.53%	3.26%	3.39%
Observations	5,514	5,514	5,514	5,514
R-squared	0.69	0.66	0.73	0.80
Game Rounds	2–4	2–4	2–4	2–4
Fixed Effects	Individual, Round	Individual, Round	Individual, Round	Individual, Round
SE Cluster	Session	Session	Session	Session

Standard errors clustered at the session level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.2 Two-limit Tobit Estimates

Table 5.2 reports two-limit Tobit estimates that account for censoring at zero and the contractual repayment obligation of 35,000 Taka. Panel A reports the latent audience effect on repayment. The latent audience effect is positive and statistically significant across all outcomes, ranging from 794 Taka in the bad yield state to 1,605 Taka in the good yield state. Relative to latent repayment without an audience, these estimates correspond to increases of approximately 3.8 to 5.3 percent. Since observed repayment is bounded, particularly with substantial mass at the full repayment limit, the latent audience effect is larger than the observed audience effect in 5.1.

Panel B reports the corresponding average marginal effects. Audience increases observed repayment by 768 Taka in the good state, 678 Taka in the normal state, and 494 Taka in the bad state, with an estimated increase of 676 Taka in expected repayment. Audience also increases the probability of full repayment by 1.1 to 2.6 percentage points across income states and reduces the probability of zero repayment by approximately 1.0 to 1.3 percentage points. For expected repayment, the corresponding changes are a 1.65 percentage-point increase in the probability of full repayment and a 0.62 percentage-point decline in the probability of zero repayment. Thus, the Tobit estimates indicate that audience shifts repayment both within the repayment bounds and toward full repayment.

Table 5.2: Audience Effect on Repayment: Two-Limit Tobit Estimates

	(1)	(2)	(3)	(4)
	<i>Good Yield Repayment</i>	<i>Normal Yield Repayment</i>	<i>Bad Yield Repayment</i>	<i>Expected Repayment</i>
Panel A: Coefficients				
Audience	1,605.15*** (413.00)	1,247.44*** (465.72)	793.76** (309.15)	862.59*** (183.06)
Constant	30,600.45*** (1,650.14)	27,374.22*** (1,514.20)	18,834.19*** (1,272.94)	22,756.09*** (916.31)
Panel B: Average Marginal Effects				
$\frac{\partial}{\partial x} E[y^* x]$ (latent, unconstrained)	1,605*** (413.0)	1,247*** (465.7)	793.8** (309.2)	862.6*** (183.1)
$\frac{\partial}{\partial x} E[y x]$ (observed, censored)	768.2*** (193.5)	678.3*** (251.1)	494.1*** (190.6)	675.5*** (142.8)
$\frac{\partial}{\partial x} \Pr[y = \bar{y} x]$ (full repayment, p.p.)	2.569*** (0.643)	2.187*** (0.810)	1.112*** (0.431)	1.653*** (0.354)
$\frac{\partial}{\partial x} \Pr[y = 0 x]$ (zero repayment, p.p.)	-1.269*** (0.337)	-1.037*** (0.394)	-1.052** (0.413)	-0.623*** (0.153)
Latent effect / latent private mean	5.25%	4.56%	4.21%	3.79%
Observations	5,514	5,514	5,514	5,514
Uncensored	2,608	2,958	3,383	4,331
Left-censored	626	581	932	281
Right-censored	2,280	1,975	1,199	902
Controls	Round, Sequence	Round, Sequence	Round, Sequence	Round, Sequence
SE Cluster	Session	Session	Session	Session

Notes: Standard errors are clustered at the session level. Censoring limits are [0,35,000] Taka. All specifications include round and game sequence controls. Panel B probability effects are reported in percentage points. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.3 Is Repayment State Contingent?

The state-specific estimates in Tables 5.1 and 5.2 show lower private repayment in worse income states and somewhat smaller point estimates of the audience effect in the bad state. To formally test whether private repayment and the audience effect vary across income states, I estimate the following specification stacking the three state contingent repayment decisions and interacting Audience with state indicators,

$$y_{igrk} = \beta \text{Audience}_g + \sum_{j \in \{N, B\}} \delta^j (\mathbf{1}\{k = j\} \times \text{Audience}_g) + \sum_{j \in \{N, B\}} \psi_j \mathbf{1}\{k = j\} + \mu_i + \lambda_r + \varepsilon_{igrk}. \quad (8)$$

where the good state in the private game is the omitted category. The state coefficients ψ_j measure differences in private repayment relative to the good state, while β denotes the audience effect in the good state. The interaction coefficients δ^j test whether the audience effect in the normal and bad states differs from that in the good state. Standard errors are clustered at the session level.

Table 5.3 reports the results. In the private games, repayment is 5,847 Taka lower in the bad state than in the good state, while the estimated difference between the normal and good states is 479

Taka. The bad-state difference is large and statistically significant, while the normal-state difference is smaller and marginally significant. In terms of the conceptual framework, this state dependence in private repayment is consistent with the marginal utility of retained income changing across income states and with the intrinsic repayment norm potentially adjusting to economic circumstances and competing needs.

In contrast, the audience effect does not vary detectably across income states. The estimated audience effect in the good state is 767 Taka. Relative to this effect, the interaction estimates are 31 Taka in the normal state and -205 Taka in the bad state, and neither is statistically significant. Thus, although private repayment changes with the income state, the additional repayment induced by audience is statistically similar across the three states.

The two-limit Tobit specification gives a similar qualitative pattern for latent repayment. The estimated latent audience effect in the good state is 1,497 Taka, while the Audience interactions are -239 Taka in the normal state and -676 Taka in the bad state. Neither interaction is statistically significant, so the Tobit estimates likewise do not reject equality of the latent audience effect across income states.

Table 5.3: State Dependence of Repayment and the Audience Effect

	(1) OLS FE	(2) Two-limit Tobit
Normal	-478.78^* (286.84)	$-1,267.60^{**}$ (564.83)
Bad	$-5,846.94^{***}$ (481.98)	$-10,059.95^{***}$ (821.83)
Audience	767.14^{***} (197.69)	$1,496.52^{***}$ (379.93)
Audience $\times$ Normal	30.83 (293.80)	-239.20 (549.39)
Audience $\times$ Bad	-204.93 (257.34)	-676.49 (449.96)
Constant	$23,108.45^{***}$ (232.77)	$29,149.24^{***}$ (1,408.39)
Observations	16,542	16,542
R-squared	0.592	0.003 (Pseudo)
Fixed Effects / Controls	Individual, Round	Game sequence, Round
Left-censored	—	2,139
Right-censored	—	5,454
Censoring Limits	—	[0, 35,000]

Notes: The omitted category is the good state in the private game. The estimated latent audience effect in the good yield state is 1,497 Taka, compared with 1,605 Taka in the state-specific Tobit estimate reported in Table 5.2. The difference is due to the stacked specification imposing a common latent error variance across income states, whereas the state specific specifications allow this variance to differ by state. Standard errors are clustered at the session level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6 Heterogeneity in Repayment and the Audience Effect

The conceptual framework discusses repayment behavior and the response to social observability to vary across borrowers. This section examines whether private repayment and the audience effect vary with educational attainment, recent shock exposure, and the geographic dispersion of households within the borrowing group. The estimates characterize reduced-form heterogeneity in repayment behavior and do not separately identify the behavioral parameters underlying these differences.

6.1 Educational Attainment

To assess whether this effect varies with education, I replace individual fixed effects round fixed effects, which allows time invariant characteristics to enter as regressors alongside their interactions with the audience indicator. Because education is absorbed by individual fixed effects in the within-subject design, the estimates in Table 6.1 rely on between-borrower variation.

The estimated audience effect among borrowers with less than Grade 9 education is positive across all income states. Borrowers with Grade 9 or higher education report substantially lower repayment in private games, whereas interaction of audience with more schooling is larger and significant in the good, normal, and expected yield states, with estimates ranging from approximately 972 to 1,341 Taka, and is imprecisely estimated in the bad yield state.

Table 6.1: Heterogeneity in the Audience Effect by Educational Attainment

	(1) <i>Good Yield Repayment</i>	(2) <i>Normal Yield Repayment</i>	(3) <i>Bad Yield Repayment</i>	(4) <i>Expected Repayment</i>
Audience	558.42*** (208.28)	629.30** (270.77)	491.84** (221.96)	566.80*** (151.45)
Above Grade 9	−4,578.45*** (1,224.95)	−5,709.18*** (1,245.29)	−6,018.54*** (1,008.02)	−5,462.77*** (1,065.68)
Audience × Above Grade 9	1,341.35** (594.83)	1,083.99* (635.74)	452.22 (541.50)	971.67** (428.26)
Constant	23,820.88*** (577.69)	23,518.04*** (575.57)	18,198.02*** (568.39)	22,012.89*** (522.80)
Observations	5,514	5,514	5,514	5,514
R-squared	0.015	0.025	0.030	0.032
Fixed Effects	Round	Round	Round	Round
SE Cluster	Session	Session	Session	Session

Standard errors clustered at the session level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.2 Flood Exposure and Household Dispersion

The conceptual framework allows a recent covariate shock to raise or lower the value of social standing, γ_i . Flood exposure may also be informally known within the VO because flooding is spatially corre-

lated across nearby households. If peers can infer which borrowers were affected, explicit audience may add less information about the reason for lower repayment. I measure household dispersion as standardized distance of households in a VO. Table 6.2 examines how the audience effect varies with flood exposure and household dispersion.

Columns (1)–(3) show that flood exposure does not increase the audience effect in any state. The differential is statistically indistinguishable from zero in the normal and bad states, while in the good state flood exposure offsets most of the audience effect, reducing it from 1,112 to approximately 317 Taka. This pattern is consistent with flood exposure providing a justification for lower repayment when repayment would otherwise be relatively high. In the normal and bad states, the income realization may already provide such a justification, leaving less scope for an audience effect of flood exposure.

Columns (4)–(6) interact flood exposure with household dispersion. Among borrowers without recent flood exposure, dispersion is unrelated to the audience effect in the good and normal states and weakly negatively associated with it in the bad state. The triple interaction is positive in all three states and statistically significant in the normal and bad states, indicating that the difference in the audience effect between flood exposed and unexposed borrowers becomes more positive as household dispersion increases. Information about flood exposure being less readily inferred when households are more geographically dispersed, increasing the role of audience. The triple interaction is small and imprecisely estimated in the good state, where the offset associated with flood exposure does not vary detectably with household dispersion. These estimates are exploratory and identify the combined reduced-form response rather than separate effects operating through γ_i or $n^R(\mathbf{x}_i)$.

Table 6.2: Heterogeneity in the Audience Effect by Flood Exposure and Household Dispersion

	Flood Exposure			Flood Exposure by Household dispersion (SD)		
	(1) Good	(2) Normal	(3) Bad	(4) Good	(5) Normal	(6) Bad
Audience	1,112.18*** (272.58)	663.46** (331.06)	769.23*** (291.28)	1,119.55*** (277.17)	666.83** (330.44)	785.22*** (293.09)
Audience $\times$ Shock	−794.72** (385.10)	309.80 (448.08)	−476.83 (375.05)	−789.59** (392.69)	397.65 (444.00)	−446.65 (371.02)
Audience $\times$ HH dispersion				−103.55 (135.57)	−47.30 (291.58)	−224.72* (135.33)
Audience $\times$ Shock $\times$ HH dispersion				238.25 (491.94)	1,030.79** (484.30)	722.64** (347.30)
Constant	23,108.45*** (97.42)	22,629.67*** (124.02)	17,261.52*** (98.56)	23,108.45*** (97.14)	22,629.67*** (122.87)	17,261.52*** (98.05)
Observations	5,514	5,514	5,514	5,514	5,514	5,514
R-squared	0.692	0.659	0.727	0.692	0.660	0.727
Fixed Effects	Indiv., Round	Indiv., Round	Indiv., Round	Indiv., Round	Indiv., Round	Indiv., Round
SE Cluster	Session	Session	Session	Session	Session	Session

Standard errors clustered at the session level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7 Interpreting the Magnitude in Lender Profitability

Microfinance institutions face substantial transaction and operating costs associated with administering and collecting many small loans (Cull et al., 2009; Rosenberg et al., 2013). Even a small increase in realized repayment can therefore have a substantial effect on lender net earnings when the lending margin is small relative to the amount intermediated. This section illustrates this relationship using a simple accounting calculation calibrated to BRAC's audited microfinance accounts.

For each taka of principal, let i_L denote the lending rate, i_B the capital cost, and c the nonfinancial operating cost. The contractual amount due is $\bar{r} = 1 + i_L$. If $d \in [0, 1]$ denotes the default share, defined as the unpaid share of the contractual amount, the lender collects $(1 - d)\bar{r}$. Net earnings per unit of principal are

$$\pi(d) = \underbrace{(1 - d)(1 + i_L)}_{\text{collections}} - \underbrace{(1 + i_B)}_{\text{capital cost}} - \underbrace{c}_{\text{operating cost}}. \quad (9)$$

Capital and operating costs do not vary with repayment in this accounting exercise, so the lender bears the cost of the default share. Under full repayment, net earnings are $\pi(0) = i_L - i_B - c$. Differentiating equation (9) gives $\partial\pi/\partial d = -(1 + i_L)$. A one percentage point increase in the default share therefore reduces net earnings by $(1 + i_L)/100$ per unit of principal. Relative to earnings under full repayment, the amplification factor is

$$\frac{-\partial\pi/\partial d}{\pi(0)} = \frac{1 + i_L}{i_L - i_B - c}. \quad (10)$$

where the break even default share is $d^* = \frac{i_L - i_B - c}{1 + i_L}$.

I calibrate these parameters using BRAC's microfinance accounts audit for fiscal year 2024–25 (BRAC, 2025) and report the calculation in terms of both principal disbursed and average principal outstanding.³ In terms of principal disbursed, the calibration uses $i_L = 0.130$, $i_B = 0.033$, and $c = 0.057$. This corresponds most closely to the one-cycle structure of equation (9). It gives $\pi(0) = 0.039$ and a break even default share of approximately 3.5 percent. For comparison, in terms of average principal outstanding, the corresponding values are $i_L = 0.227$, $i_B = 0.058$, and $c = 0.100$, with $\pi(0) = 0.069$ and a break even default share of approximately 5.6 percent. The latter is directly comparable to portfolio yields and expense ratios commonly reported for microfinance institutions. Table 7.1 reports net earnings under alternative default shares using both calibrations. In the experiment, an increase in observed repayment of Δr corresponds to a reduction in the default share of the contractual obligation of $\Delta d = -\Delta r/\bar{r}$. For example, the estimated increase in observed expected repayment of 675.5 Taka against the debt obligation of 35,000 Taka corresponds to a 1.93 percentage-point reduction in the de-

³BRAC reports microfinance service-charge income of 93.7 billion Taka, financial expenditure of approximately 24.0 billion Taka, and total operating expenditure of approximately 65.3 billion Taka in fiscal year 2024–25. Nonfinancial operating expenditure is therefore approximately 41.3 billion Taka. In terms of total principal disbursed of 721.1 billion Taka, these give $i_L = 0.130$, $i_B = 0.033$, and $c = 0.057$. Average principal outstanding is approximately 413.5 billion Taka, calculated from 397.0 billion Taka at the beginning of the year and 429.9 billion Taka at the end of the year. In terms of average principal outstanding, $i_L = 0.227$, $i_B = 0.058$, and $c = 0.100$. Loss provisions are not included in c because repayment losses are represented directly by the default share d in equation (9).

Table 7.1: Net Earnings under Alternative Default Shares

Default share d	Principal Disbursed		Average Principal Outstanding	
	Net earnings $\pi(d)$	Loss (% of $\pi(0)$)	Net earnings $\pi(d)$	Loss (% of $\pi(0)$)
0.0%	0.0394	0	0.0688	0
0.5%	0.0338	14	0.0626	9
1.0%	0.0281	29	0.0565	18
1.93% ^a	0.0176	55	0.0451	34
2.0%	0.0168	57	0.0442	36
3.0%	0.0055	86	0.0320	54
3.5%	0.0000	100	0.0258	62
5.6%	-0.0239	161	0.0000	100

Notes. Calculated from equation (9) using parameters calibrated to BRAC's audited microfinance accounts for fiscal year 2024–25 (BRAC, 2025). The first calibration is expressed in terms of principal disbursed; the second is expressed in terms of average principal outstanding. ^aThe 1.93 percent row corresponds to the magnitude of the experimental-equivalent reduction in the default share discussed in the text. Because net earnings are linear in d , the earnings loss shown at this default share is also the earnings gain associated with a 1.93 percentage-point reduction in the default share.

fault share of the experimental obligation. If an equivalent reduction in the default share occurred on an otherwise comparable field loan, the BRAC calibrations imply an increase in lender net earnings equivalent to approximately 55 percent of full-repayment earnings in terms of principal disbursed and 34 percent in terms of average principal outstanding.

8 Conclusion

Microfinance institutions retain group-based repayment practices after moving from joint to individual liability that make delinquency observable to peers. This paper examines whether this remaining socially embedded feature affects repayment. Using within-subject variation in whether default is observable to peers, that is, whether the repayment decision is made before an audience, I find that social observability raises observed repayment by 3.3 to 3.5 percent and latent repayment by 3.8 to 5.3 percent. Since the experiment holds monetary incentives fixed across conditions and removes consequences for future credit access, these estimates isolate the audience effect on repayment from other contractual enforcement mechanisms. The shift to individual liability therefore removes shared financial responsibility without necessarily eliminating socially embedded repayment incentives.

Repayment without an audience is state dependent. Borrowers repay between one-half and two-thirds of the debt obligation, with repayment falling in the adverse income state even though full repayment remains feasible. In contrast, the audience effect does not vary detectably across income states. This finding is consistent with the interpretation that private repayment adjusts to realized economic circumstances, while social observability induces an additional repayment reflecting reputational motivation across income states.

An important feature of the experiment is that peers would observe the borrower's default amount but not the income state in which that decision was made. The absence of detectable state dependence in the average audience effect is consistent with this information structure. The exploratory hetero-

geneity results nevertheless suggest that peers may draw on information beyond what the lender explicitly reveals. Recent flood exposure offsets much of the audience effect in the good state, consistent with prior hardship providing a plausible justification for lower repayment even when current income is favorable. In the normal and bad states, the interaction between flood exposure and the audience effect becomes more positive as households within the borrowing group are more geographically dispersed. One possible interpretation is that audience matters more when peers have less direct information about one another's circumstances. These estimates do not separately identify this informational mechanism, but they suggest that the effect of observable default may depend partly on what peers can otherwise infer about the circumstances surrounding repayment.

The repayment gain is small relative to the total debt obligation but can matter for lenders when lending margins are thin. The estimated increase in observed expected repayment of 675.5 Taka corresponds to a 1.93 percentage-point reduction in the default share. If a comparable reduction in field default occurred on an otherwise similar loan, the BRAC accounting calibration implies a profit gain equivalent to approximately 34–55 percent of the zero-default profit margin. This calculation illustrates why relatively small changes in repayment can have meaningful implications for lender profitability.

This profitability calculation does not, however, establish that strengthening social observability is welfare improving. It reflects a potential gain to the lender but does not account for nonpecuniary costs to borrowers. If reputational motivation makes borrowing more costly, socially observable repayment practices could affect credit demand even where microfinance relaxes conventional collateral requirements (Stiglitz and Weiss, 1981; Boucher et al., 2008). This is relevant in settings where households rely on the same social relationships for informal support and risk sharing.

The repeated repayment decisions also provide a basis for studying borrower level heterogeneity in repayment behavior and responses to social observability beyond the experiment. Choices are highly consistent within borrowers across states and rounds, suggesting persistent borrower-specific variation in repayment behavior. Future work can examine whether borrower-level measures constructed from private and socially observable repayment decisions predict borrowing, credit demand, and demand for risk-management products in the field (Castillo et al., 2018). A related extension would vary not only whether repayment is observable, but also what information about the borrower's circumstances is available to the audience. Such variation would help distinguish the effect of making default visible from the effect of making the circumstances surrounding default interpretable to peers.

References

- AKERLOF, G. A. (1980): "A theory of social custom, of which unemployment may be one consequence," *The quarterly journal of economics*, 94, 749–775.
- ANDREONI, J. AND B. D. BERNHEIM (2009): "Social image and the 50–50 norm: A theoretical and experimental analysis of audience effects," *Econometrica*, 77, 1607–1636.
- ARMENDÁRIZ, B. AND J. MORDUCH (2000): "Microfinance beyond group lending," *Economics of transition*, 8, 401–420.
- (2010): *The economics of microfinance*, The MIT Press.
- BALBONI, C., O. BANDIERA, R. BURGESS, M. GHATAK, AND A. HEIL (2022): "Why do people stay poor?" *The Quarterly Journal of Economics*, 137, 785–844.
- BÉNABOU, R. AND J. TIROLE (2006): "Incentives and prosocial behavior," *American Economic Review*, 96, 1652–1678.
- BESLEY, T. AND S. COATE (1995): "Group lending, repayment incentives and social collateral," *Journal of Development Economics*, 46, 1–18.
- BOUCHER, S. R., M. R. CARTER, AND C. GUIRKINGER (2008): "Risk rationing and wealth effects in credit markets: Theory and implications for agricultural development," *American Journal of Agricultural Economics*, 90, 409–423.
- BRAC (2025): "Annual Audit Report 2024–2025," <https://www.brac.net/documents/814/Annual-Audit-Report-2024-25.pdf>, microfinance programme annexures and Notes 14 and 31.
- CASTILLO, M., J. L. JORDAN, AND R. PETRIE (2018): "Children's rationality, risk attitudes and field behavior," *European economic review*, 102, 62–81.
- CIALDINI, R. B. AND N. J. GOLDSTEIN (2004): "Social influence: Compliance and conformity," *Annu. Rev. Psychol.*, 55, 591–621.
- CIALDINI, R. B., C. A. KALLGREN, AND R. R. RENO (1991): "A focus theory of normative conduct: A theoretical refinement and reevaluation of the role of norms in human behavior," in *Advances in experimental social psychology*, Elsevier, vol. 24, 201–234.
- CULL, R., A. DEMIRGÜÇ-KUNT, AND J. MORDUCH (2009): "Microfinance meets the market," *Journal of Economic perspectives*, 23, 167–192.
- DANA, J., R. A. WEBER, AND J. X. KUANG (2007): "Exploiting moral wiggle room: experiments demonstrating an illusory preference for fairness," *Economic Theory*, 33, 67–80.

- DHAMI, S., J. ARSHAD, AND A. AL-NOWAIHI (2022): "Psychological and social motivations in microfinance contracts: Theory and evidence," *Journal of Development Economics*, 158, 102912.
- FIELD, E. AND R. PANDE (2008): "Repayment frequency and default in microfinance: Evidence from India," *Journal of the European Economic Association*, 6, 501–509.
- FLATNES, J. E. AND M. R. CARTER (2019): "A little skin in the game: Reducing moral hazard in joint liability lending through a mandatory collateral requirement," *Journal of Economic Behavior & Organization*, 164, 199–214.
- GHATAK, M. (2001): "Screening by the Company You Keep: Joint Liability Lending and the Peer Selection Effect," *The Economic Journal*, 110, 601–631.
- GINÉ, X., J. GOLDBERG, AND D. YANG (2010a): "Identification strategy: A field experiment on dynamic incentives in rural credit markets," *Available at SSRN 1633517*.
- GINÉ, X., P. JAKIELA, D. KARLAN, AND J. MORDUCH (2010b): "Microfinance games," *American Economic Journal: Applied Economics*, 2, 60–95.
- GINÉ, X. AND D. KARLAN (2014): "Group versus individual liability: Long term evidence from Philippine microcredit lending groups," *Journal of Development Economics*, 107, 65–83.
- HOENIG, J. M. AND D. M. HEISEY (2001): "The abuse of power: the pervasive fallacy of power calculations for data analysis," *The American Statistician*, 55, 19–24.
- KIMBROUGH, E. O. AND A. VOSTROKNUTOV (2016): "Norms make preferences social," *Journal of the European Economic Association*, 14, 608–638.
- LIST, J. A. (2025): "The Experimentalist Looks Within: Toward an Understanding of Within-Subject Experimental Designs," NBER Working Paper 33456, National Bureau of Economic Research.
- MADAJEWICZ, M. (2011): "Joint liability versus individual liability in credit contracts," *Journal of Economic Behavior & Organization*, 77, 107–123.
- MCINTOSH, C. (2008): "Estimating treatment effects from spatial policy experiments: an application to Ugandan microfinance," *The Review of Economics and Statistics*, 90, 15–28.
- ROSENBERG, R., S. GAUL, W. FORD, AND O. TOMILOVA (2013): "Microcredit Interest Rates and Their Determinants: 2004–2011," in *Microfinance 3.0: Reconciling Sustainability with Social Outreach and Responsible Delivery*, ed. by D. Köhn, Berlin, Heidelberg: Springer, 69–104.
- STIGLITZ, J. E. (1990): "Peer monitoring and credit markets," *The world bank economic review*, 4, 351–366.
- STIGLITZ, J. E. AND A. WEISS (1981): "Credit rationing in markets with imperfect information," *The American economic review*, 71, 393–410.

A Appendix Figures and Tables

Figure A.1: Study Site

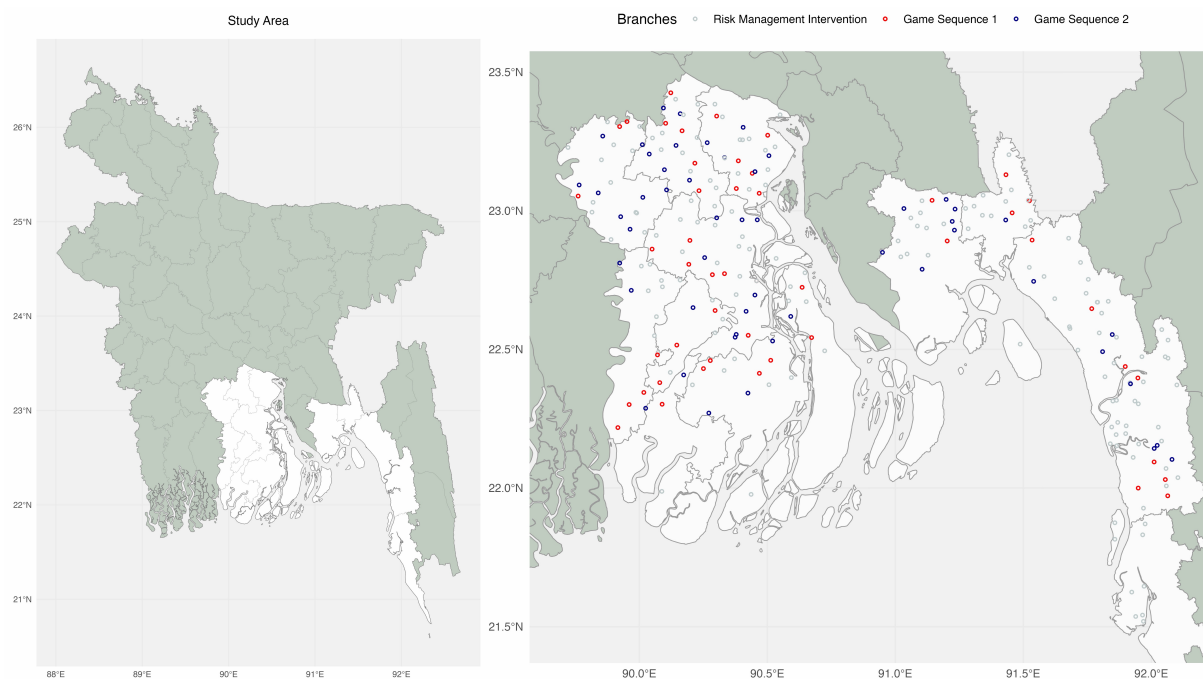


Figure A.2: Randomization of Game Design

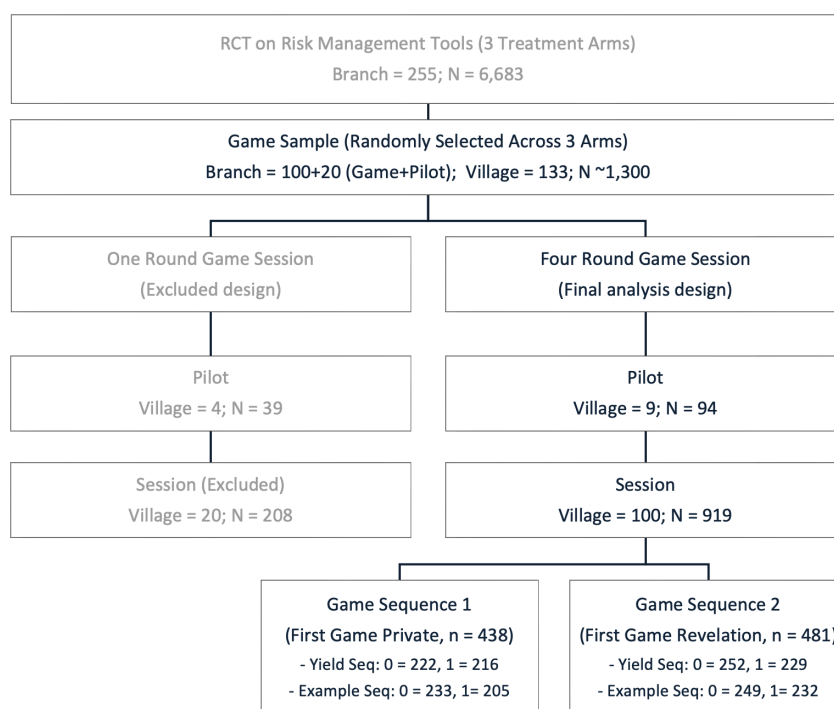
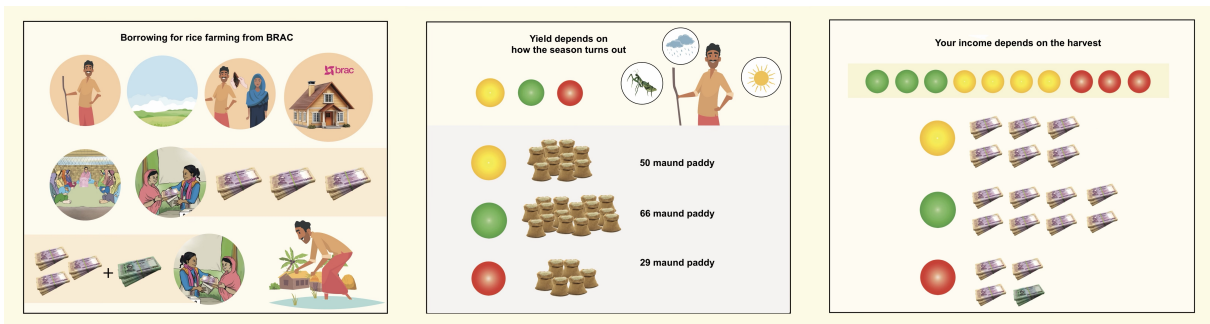


Figure A.3: Actual village meeting and experimental session settings



Notes: The left panel shows a BRAC village organization meeting, illustrating the group setting in which microfinance activities take place. The right panel shows an experimental session conducted at a village meeting location. The left-panel photograph is from BRAC, ([url](#)).

Figure A.4: Visual aids for the experimental scenario



Notes: The visual aids were used to explain the rice-farming loan, uncertainty in harvest outcomes, and the corresponding income states. Illustrations of Bangladeshi Taka were used so that participants could follow the monetary amounts without relying on literacy.

Figure A.5: Decision sheet and board to reveal default for the audience game

Income Principal + Interest				
Decision	Repay	Default	Remaining Money	Actual Earning
	35,000	0	45,000	450
	30,000	5,000	50,000	500
	25,000	10,000	55,000	550
	20,000	15,000	60,000	600
	15,000	20,000	65,000	650
	10,000	25,000	70,000	700
	5,000	30,000	75,000	750
	0	35,000	80,000	800

Name of the Borrower	Amount Defaulted
Farhana Akter	
Shyamoli Rani Das	
Mousumi Akter	
Nipa Begum	
Sharmin Sultana	
Monira Mondol	
Tania Begum	
Nahida Parvin	
Rukaiya Haque	
Sanjida Siddika	
Purnima Sarker	

Notes: The left panel shows the repayment decision sheet for the good income state. Participants selected a repayment amount from the available choices, with the corresponding default, remaining income, and experimental earnings shown for each choice. Analogous decision sheets were used for the normal and bad income states, with amounts adjusted to the corresponding income levels. The right panel shows the board displayed in the audience game to indicate that the default amount from the randomly selected audience game would be revealed to members of the borrowing group.

Figure A.6: Repayment distribution across conditions and income states (round 2-4)

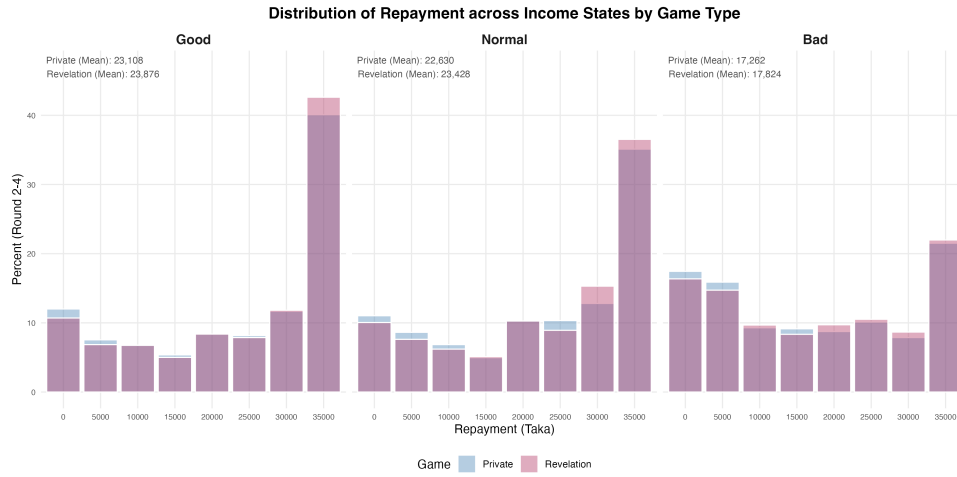


Table A.1: Descriptive statistics of repayment decisions, by round, audience, and income state

Round	Observability	Income state	N	Mean	S.D.	Min	p25	p50	p75	Max
1	Private	Good	919	27,149	11,584	0	20,000	35,000	35,000	35,000
		Normal	919	27,873	10,920	0	25,000	35,000	35,000	35,000
		Bad	919	21,616	12,945	0	10,000	25,000	35,000	35,000
	Audience	Good	919	27,998	10,965	0	25,000	35,000	35,000	35,000
		Normal	919	28,351	10,689	0	25,000	35,000	35,000	35,000
		Bad	919	22,378	12,848	0	10,000	25,000	35,000	35,000
2	Private	Good	919	23,781	12,699	0	15,000	30,000	35,000	35,000
		Normal	919	23,058	12,586	0	10,000	25,000	35,000	35,000
		Bad	919	18,112	13,181	0	5,000	20,000	30,000	35,000
	Audience	Good	919	24,374	12,488	0	15,000	30,000	35,000	35,000
		Normal	919	23,808	12,357	0	15,000	30,000	35,000	35,000
		Bad	919	18,939	12,784	0	5,000	20,000	30,000	35,000
3	Private	Good	919	22,894	13,067	0	10,000	30,000	35,000	35,000
		Normal	919	22,884	12,551	0	10,000	25,000	35,000	35,000
		Bad	919	17,035	12,899	0	5,000	15,000	30,000	35,000
	Audience	Good	919	23,993	12,598	0	15,000	30,000	35,000	35,000
		Normal	919	23,738	12,045	0	15,000	30,000	35,000	35,000
		Bad	919	17,845	13,046	0	5,000	20,000	30,000	35,000
4	Private	Good	919	22,650	13,035	0	10,000	30,000	35,000	35,000
		Normal	919	21,948	12,696	0	10,000	25,000	35,000	35,000
		Bad	919	16,638	12,824	0	5,000	15,000	30,000	35,000
	Audience	Good	919	23,259	12,957	0	10,000	30,000	35,000	35,000
		Normal	919	22,737	12,663	0	10,000	30,000	35,000	35,000
		Bad	919	16,687	12,784	0	5,000	15,000	30,000	35,000

Notes: Repayment amounts are in Bangladeshi Taka; the maximum feasible repayment is 35,000. Means and standard deviations are rounded to the nearest Taka.

B Optimal Repayment

An interior optimum $r^*(s)$ satisfies

$$u'(y(\theta_i) - r^*(s)) = \alpha_i N'(r^*(s) - n^I(\theta_i)) \mathbf{1}\{r^*(s) < n^I(\theta_i)\} + s\gamma_i N'(r^*(s) - n^R(\mathbf{x}_i)) \mathbf{1}\{r^*(s) < n^R(\mathbf{x}_i)\}.$$

Under concavity, the optimum is $r^*(s) = 0$ if $U_r(0; s, \theta_i, \mathbf{x}_i) \leq 0$ and $r^*(s) = \bar{r}$ if $U_r(\bar{r}; s, \theta_i, \mathbf{x}_i) \geq 0$. Otherwise, $r^*(s) \in (0, \bar{r})$ is the unique solution to the first-order condition above.

The second-order condition is

$$u''(y(\theta_i) - r) + \alpha_i N''(r - n^I(\theta_i)) \mathbf{1}\{r < n^I(\theta_i)\} + s\gamma_i N''(r - n^R(\mathbf{x}_i)) \mathbf{1}\{r < n^R(\mathbf{x}_i)\} < 0.$$

B.1 Optimal Repayment under Specific Functional Forms

To illustrate the model, let

$$u(c) = \sqrt{c}, \quad N(z) = \kappa \min\{z, 0\} - \frac{1}{2}[\min\{z, 0\}]^2, \quad y(\theta_i) = \theta_i \in \{\theta^G, \theta^N, \theta^B\},$$

To simplify, I set both repayment standards equal to the contractual obligation, $n^I(\theta_i) = n^R(\mathbf{x}_i) = \bar{r}$. For $r < \bar{r}$, utility is,

$$U(r; s, \theta_i) = \sqrt{\theta_i - r} - (\alpha_i + s\gamma_i) \left[\kappa(\bar{r} - r) + \frac{1}{2}(\bar{r} - r)^2 \right],$$

which is strictly concave, since $U'' = -\frac{1}{4}(\theta_i - r)^{-3/2} - (\alpha_i + s\gamma_i) < 0$, so the optimum is unique.

The optimum $r^*(s)$ is

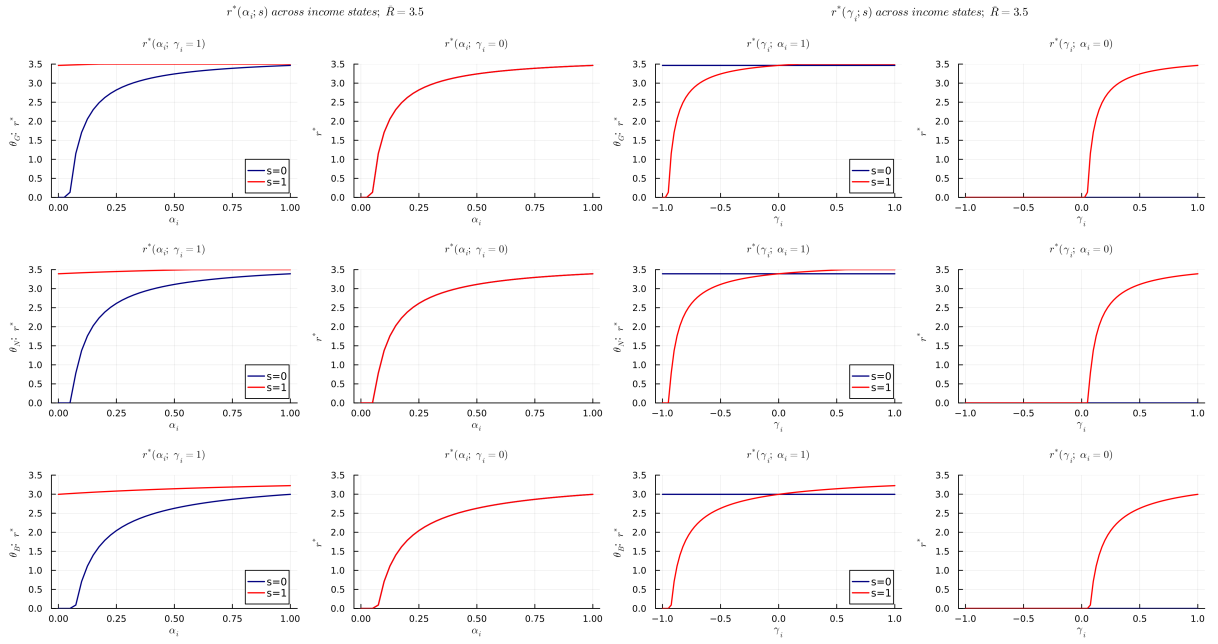
$$r^*(s) \in \begin{cases} 0 & \text{when } (\alpha_i + s\gamma_i)(\kappa + \bar{r}) \leq \frac{1}{2\sqrt{\theta_i}}, \\ (0, \bar{r}) & \text{when } \frac{1}{2\sqrt{\theta_i - r^*}} = (\alpha_i + s\gamma_i)(\kappa + \bar{r} - r^*), \\ \bar{r} & \text{when } (\alpha_i + s\gamma_i)\kappa \geq \frac{1}{2\sqrt{\theta_i - \bar{r}}}. \end{cases}$$

The interior condition is a cubic in r^* , solved numerically. The full repayment occurs as $\kappa > 0$ implies a positive marginal norm penalty at $r = \bar{r}$. Since the two norms enter only through the sum $\alpha_i + s\gamma_i$, which is larger under $s = 1$, and $r^*(s)$ is increasing in that sum, observability weakly raises repayment, $r^*(1) \geq r^*(0)$ for $\gamma_i \geq 0$, with equality possible when both choices remain at the same corner.

Table B.1: Parameterization for the comparative-statics figures

	Value	Description
$\bar{r}$	3.5	Contractual obligation, $r \in [0, \bar{r}]$
$(\theta^G, \theta^N, \theta^B)$	(8.0, 6.0, 3.5)	Good / normal / bad income realization
s	$\{0, 1\}$	Social Observability
α_i	$[0, 1]$	Intrinsic motivation heterogeneity
γ_i	$[-1, 1]$	Reputational motivation heterogeneity
κ	0.2	Penalty slope at benchmark

Figure B.1: Optimal repayment in α_i and γ_i space



C Within-Subject Design Diagnostics

C.1 Consistency of Repayment Choices Across States and Rounds

I first assess whether repayment choices exhibit persistent borrower-specific variation within each experimental condition rather than idiosyncratic variation across repeated decisions. Following Castillo et al. (2018), I compare the observed consistency of repayment choices with Monte Carlo distributions generated under random choice.

The design permits two consistency measures. Panel A of Table C.1 treats the good, normal, and bad income states as three items and estimates a separate coefficient for each round and experimental condition. This measure assesses whether borrowers who repay relatively more in one income state also tend to repay relatively more in the other states. Panel B treats analysis rounds two through four as the three items while holding the income state and experimental condition fixed. This measure assesses whether borrowers maintain their relative repayment behavior across repeated decisions.

Each scale contains three items. The standardized Cronbach's alpha reported in the table is

$$\text{Cronbach's alpha} = \frac{\rho_{12} + \rho_{13} + \rho_{23}}{1 + \frac{2}{3}(\rho_{12} + \rho_{13} + \rho_{23})},$$

where ρ_{12} , ρ_{13} , and ρ_{23} denote the pairwise correlations among the three repayment choices used as items in the corresponding scale. A higher coefficient indicates greater persistence in between-borrower differences across the three items. The coefficient therefore measures consistency in borrowers' relative repayment behavior rather than constancy of repayment amounts.

The coefficients in Panel A range from 0.791 to 0.809, indicating substantial consistency in borrowers' relative repayment choices across income states. The coefficients in Panel B range from 0.810 to 0.869, indicating similarly strong consistency across rounds. The coefficients are similar in magnitude across the private and audience games, with no apparent reduction in consistency under social observability. They also remain similar across rounds despite the decline in average repayment over the session. Across-round consistency is numerically highest in the bad state, with coefficients of 0.849 in the private game and 0.869 in the audience game.

I compare the across-round coefficients in Panel B with two random-choice Monte Carlo distributions. The first is generated by drawing each repayment independently from the eight-point feasible repayment grid. The second is generated by independently resampling repayment from the observed distribution in each round within the corresponding income state and experimental condition. The latter preserves the round-specific marginal distributions of repayment while removing the borrower-specific association across rounds. Under both procedures, the three repayment choices are drawn independently, so their population correlations are zero and any sample correlation arises from finite-

sample variation.

The resulting Cronbach’s alpha distributions are centered near zero. The 95th percentile is 0.085 under random draws from the uniform repayment grid and ranges from 0.085 to 0.095 under empirical-marginal resampling across the six state-by-condition cells. None of the 1,000 Monte Carlo draws reaches the corresponding observed coefficient, yielding a Monte Carlo p -value of 0.001 in each cell. The observed across-round coefficients are therefore substantially larger than those generated by random choice, including when the observed round-specific repayment distributions are preserved.

These results are consistent with persistent borrower-level heterogeneity in repayment behavior across states and rounds. They do not imply that repayment remains constant in levels over time or that borrower-specific audience effects are stable across rounds. The coefficients measure persistence in repayment levels within each condition, not persistence in the within-borrower difference between the audience and private conditions. Changes in repayment levels and in the average audience effect across rounds are examined separately in Appendix C.2.

Table C.1: Consistency of repayment choices across states and rounds

	<i>Private</i>	<i>Audience</i>
<i>Panel A. Across income states</i>		
Round 2	0.807	0.796
Round 3	0.798	0.799
Round 4	0.791	0.809
<i>Panel B. Across analysis rounds</i>		
Good yield	0.841	0.832
Normal yield	0.822	0.810
Bad yield	0.849	0.869
Items per scale	3	3
Borrowers	919	919
<i>Random-choice Monte Carlo comparison for Panel B</i>		
Uniform grid, 95th percentile	0.085	
Empirical marginal, 95th percentile	0.085–0.095	

Notes. Entries report standardized Cronbach’s alpha coefficients. Panel A uses the three income states as items, while Panel B uses analysis rounds two through four as items. Private and audience choices are evaluated separately. The uniform-grid comparison draws repayment independently from the eight-point choice set. The empirical-marginal comparison independently resamples from the observed round-specific repayment distributions. Both use 1,000 Monte Carlo draws of 919 borrowers. The empirical-marginal entry reports the range of 95th percentiles across the six state-by-condition cells.

C.2 Pooling Across Rounds

Because each borrower is observed under both experimental conditions in each of three rounds, I examine whether repayment changes systematically over the session and whether the audience effect varies across rounds or depends on treatment order. Following List (2025), this section examines round trends in repayment, variation in the audience effect across rounds, and asymmetry by treatment order. Repayment levels decline over the session, but the audience effect does not vary detectably

across rounds and there is no detectable order asymmetry. These patterns support reporting a pooled audience effect across rounds 2–4 with round fixed effects, while recognizing that the estimates are not sufficiently precise to rule out moderate round-specific variation.

C.2.1 Round Trends in Repayment

Figure C.1 plots mean repayment by round for each income state and experimental condition. Repayment declines from round 2 to round 4 in both the private and audience games.

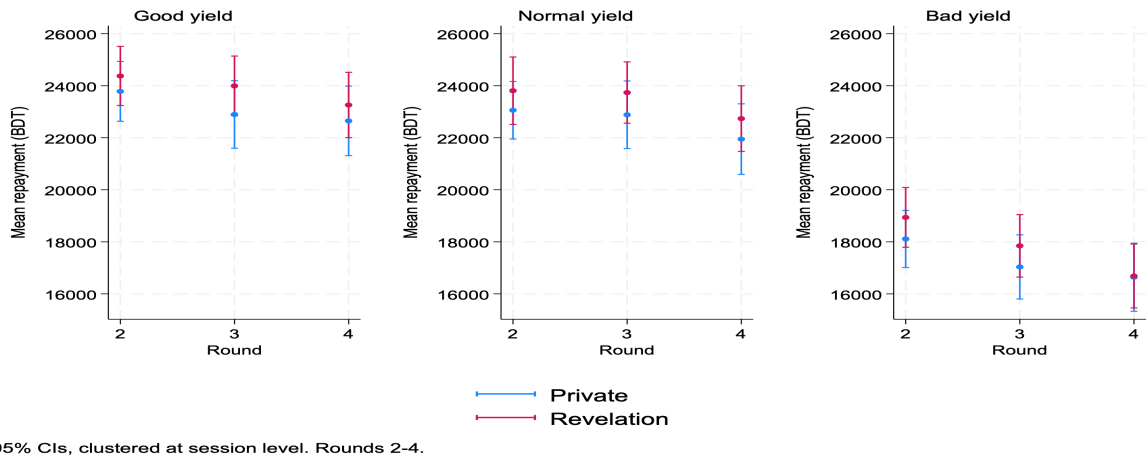
To quantify this pattern, I estimate

$$R_{it} = \mu_i + \sum_{t=3}^4 \delta_t \mathbb{1}[\text{round} = t] + \beta \text{Audience}_{it} + \varepsilon_{it}, \quad (11)$$

separately by income state, where R_{it} denotes borrower i 's repayment choice in round t and μ_i is a borrower fixed effect. Standard errors are clustered at the session level. Round 2 and the private condition are the omitted categories.

Panel A of Table C.2 shows that the round coefficients are jointly different from zero in all three income states. Relative to round 2, repayment in round 4 is lower by 1,863 Taka in the bad state, 1,091 Taka in the normal state, and 1,124 Taka in the good state. The decline therefore reflects a general change in repayment levels over the session rather than a change specific to the audience game. Because round is perfectly correlated with position in the session, the data do not distinguish learning, fatigue, or other time-related explanations for this decline.

Figure C.1: Repayment Levels by Round



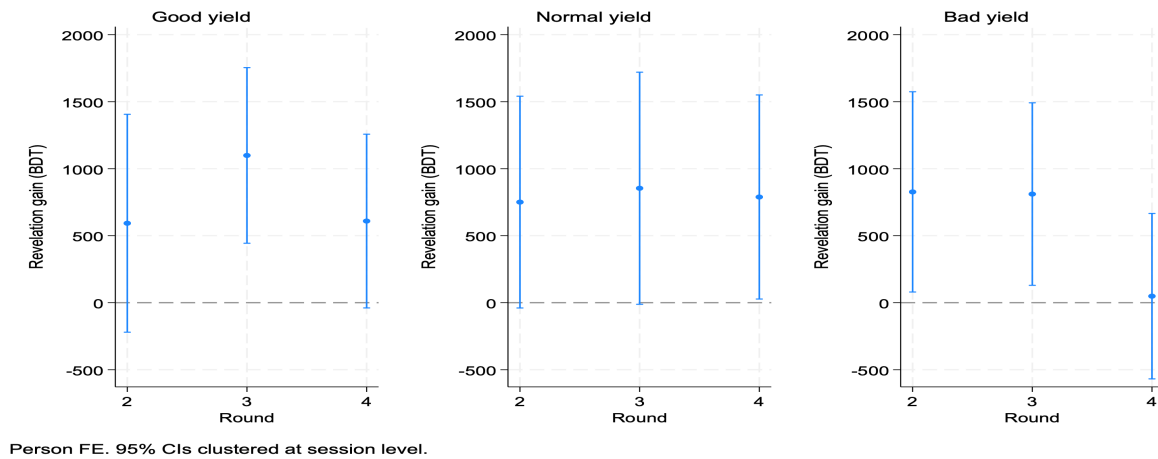
C.2.2 Stability of the Audience Effect Across Rounds

I next examine whether the audience effect varies systematically across rounds. Panel B of Table C.2 augments equation (11) with round-by-audience interactions. The interactions are jointly indistinguishable from zero in all three income states, with $p = 0.183$ in the bad state, $p = 0.983$ in the normal state,

and $p = 0.445$ in the good state.

Figure C.2 plots the implied audience effect separately by round. The point estimates do not show a systematic pattern across rounds, although they are imprecisely estimated. In particular, the interaction standard errors range from approximately 512 to 581 Taka, relative to pooled audience effects of 562 to 798 Taka. The estimates therefore provide no detectable evidence of systematic variation in the audience effect across rounds, but they cannot rule out economically meaningful round-specific differences, particularly in the bad income state.

Figure C.2: Audience Effect by Round



C.2.3 Order Effects

A separate concern in a within-subject design is that exposure to one experimental condition may affect behavior in the other. Because the order of the private and audience games was randomized at the session level, I test whether the audience effect differs between sessions in which the audience game was presented first and those in which the private game was presented first. Panel C reports these specifications without borrower fixed effects.

Panel C of Table C.2 shows no detectable interaction between the audience condition and treatment order in any income state or in the pooled specification. In the pooled model, the interaction is -170 Taka with $p = 0.554$. The state-specific estimates are also statistically indistinguishable from zero. As with the round interactions, these estimates are not sufficiently precise to exclude moderate order effects, but there is no systematic evidence that the audience effect differs depending on which condition borrowers encounter first.

Taken together, these diagnostics support pooling rounds 2–4 while controlling for round fixed effects. Repayment levels change over the session, but the round-by-audience interactions provide no detectable evidence that the audience effect changes systematically with round. The randomized order comparison similarly provides no detectable evidence that the audience effect differs by treatment

order. I therefore include round fixed effects throughout the main specifications and interpret the pooled coefficient as the average audience effect across rounds 2–4.

Table C.2: Round Stability and Order Effects

	Good yield	Normal yield	Bad yield	Pooled
<i>Panel A. Round trends, borrower fixed effects</i>				
Round 3	−633.84** (302.72)	−122.42 (308.46)	−1085.42*** (309.78)	
Round 4	−1123.50*** (375.14)	−1090.86*** (344.79)	−1863.44*** (350.63)	
Audience	767.14*** (197.70)	797.97*** (248.96)	562.21*** (198.45)	
<i>F</i> , round effects jointly zero	4.54	7.00	14.15	
<i>p</i> -value	0.013	0.001	0.000	
<i>Panel B. Round by audience, borrower fixed effects</i>				
Round 3 × Audience	505.98 (569.44)	103.37 (581.31)	−16.32 (512.17)	
Round 4 × Audience	16.32 (573.74)	38.08 (580.66)	−778.02 (515.50)	
<i>F</i> , interactions jointly zero	0.82	0.02	1.73	
<i>p</i> -value	0.445	0.983	0.183	
<i>Panel C. Order effects, no borrower fixed effects</i>				
Audience × Audience first	498.76 (391.39)	−518.32 (496.38)	−488.94 (393.16)	−169.50 (285.07)
<i>F</i> , interaction zero	1.62	1.09	1.55	0.35
<i>p</i> -value	0.206	0.299	0.217	0.554
Observations	5,514	5,514	5,514	16,542
Borrowers	919	919	919	919
Session clusters	100	100	100	100

Notes. The dependent variable is repayment in Taka against an obligation of 35,000 Taka. Standard errors in parentheses are clustered at the session level. Round 2 and the private condition are the omitted categories. Panels A and B include borrower fixed effects and are estimated separately by income state. Panel C tests whether the audience effect differs when the audience game is presented first. Panel C is estimated without borrower fixed effects. The pooled specification includes income-state indicators. Stars denote *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$.

C.3 Precision of Estimates

The stacked specifications provide no detectable evidence that the audience effect differs across income states. Because failure to reject differences does not establish equality, particularly when the interaction terms are imprecisely estimated, I summarize the precision of these estimates using approximate minimum detectable effects. I report these rather than observed power, which is a monotone transformation of the realized *p*-value and provides little additional information once the estimate is observed (Hoenig and Heisey, 2001). Table C.3 reports approximate minimum detectable effects for the coefficients of interest, calculated as

$$\text{MDE} = (z_{0.975} + z_{0.80}) \hat{\text{se}},$$

for a two-sided test at the 5 percent level with 80 percent power.

Panel A reports the fixed effects estimates on observed repayment. The audience effect is estimated more precisely than the state interactions. Its minimum detectable effect is 554 Taka, or approximately 2.4 percent of private-game repayment in the good income state, compared with an estimated effect of 767 Taka. The state interactions are less precisely estimated. The minimum detectable difference is approximately 823 Taka for the normal-state interaction and 721 Taka for the bad-state interaction.

The 95 percent confidence interval for the bad-state interaction ranges from -716 to 306 Taka. The estimate therefore remains compatible with differences of several hundred Taka across income states. The estimates provide no detectable evidence of state heterogeneity in the audience effect, but they do not establish that the effect is identical across states.

Panel B reports the corresponding precision for the two-limit Tobit estimates of latent repayment. The same pattern holds. The audience coefficient is estimated more precisely than the state interactions, while the confidence intervals around the interaction terms remain wide. Because the Tobit coefficients refer to latent repayment, their levels and minimum detectable effects are not directly comparable with the fixed effects estimates on observed repayment.

The round-by-audience interactions in Section C.2 are less precise still. Using the same calculation, their minimum detectable effects range from approximately 1,435 to 1,629 Taka, compared with pooled audience effects of 562 to 798 Taka. The data therefore cannot establish that the audience effect is constant across rounds. I instead interpret the pooled specification as estimating the average audience effect across rounds 2–4 while recognizing that moderate round-specific variation cannot be ruled out.

Table C.3: Precision of the Audience and State-Interaction Estimates

	Estimate	S.E.	MDE	95% CI
<i>Panel A. OLS fixed effects</i>				
Audience	767.1	197.7	553.9	—
Normal $\times$ Audience	30.8	293.8	823.1	[-552 , 614]
Bad $\times$ Audience	-204.9	257.3	721.0	[-716 , 306]
<i>Panel B. Two-limit Tobit</i>				
Audience	1,496.5	379.9	1,064.4	—
Normal $\times$ Audience	-239.2	549.4	1,539.2	[-1 , 316 , 838]
Bad $\times$ Audience	-676.5	450.0	1,260.6	[-1 , 558 , 205]

Notes. The minimum detectable effect is approximated by $(z_{0.975} + z_{0.80}) \times \widehat{se}$ and gives the approximate coefficient magnitude associated with 80 percent power in a two-sided test at the 5 percent level, conditional on the estimated standard error. Standard errors are clustered at the session level. Estimates, standard errors, and confidence intervals are taken from Table 5.3. The Tobit estimates refer to latent repayment and are not directly comparable with the OLS estimates on observed repayment.

D Robustness to Clustering and Final-Round Exclusion

I examine two robustness checks for the main fixed effects estimates. First, the preferred specifications cluster standard errors at the session level to allow for correlation across decisions made within the same experimental session. Panel A in Table D.1 instead clusters standard errors at the borrower level. The standard errors and statistical significance are similar under the alternative clustering level, and the estimated audience effects are unchanged.

Second, I restrict the sample to rounds 2 and 3 to examine whether the main estimates are sensitive to behavior in the final experimental round. Panel B in Table D.1 shows that the audience effect remains positive and statistically significant across all income states and for expected repayment. The estimates are similar in magnitude to, and in some cases somewhat larger than, those obtained using rounds 2–4. The main results therefore do not appear to be driven by inclusion of the final round.

Table D.1: Robustness of the Audience Effect

	(1) <i>Good Yield Repayment</i>	(2) <i>Normal Yield Repayment</i>	(3) <i>Bad Yield Repayment</i>	(4) <i>Expected Repayment</i>
<i>Panel A. Borrower-clustered standard errors</i>				
Audience	767.14*** (211.20)	797.97*** (221.73)	562.21*** (198.31)	717.99*** (145.82)
Constant	23,108.45*** (105.60)	22,629.67*** (110.86)	17,261.52*** (99.15)	21,162.86*** (72.91)
Observations	5,514	5,514	5,514	5,514
Game Rounds	2–4	2–4	2–4	2–4
SE Cluster	Borrower	Borrower	Borrower	Borrower
<i>Panel B. Excluding the final round</i>				
Audience	846.03*** (246.57)	802.50** (307.10)	818.82*** (259.55)	820.46*** (167.99)
Constant	23,337.87*** (123.29)	22,970.62*** (153.55)	17,573.45*** (129.78)	21,461.64*** (84.00)
Observations	3,676	3,676	3,676	3,676
Game Rounds	2–3	2–3	2–3	2–3
SE Cluster	Session	Session	Session	Session
Fixed Effects	Individual, Round	Individual, Round	Individual, Round	Individual, Round

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

E Monte Carlo Assessment of Censored Outcome Estimators

I use Monte Carlo simulations to assess the interpretation of the fixed effects and two-limit Tobit estimates when repayment is censored at the experimental bounds. The exercise distinguishes the audience effect on observed repayment, $\beta^O = E[y_{ir}^A - y_{ir}^P]$, from the audience effect on latent repayment, β . I calibrate the simulations to the Normal-state repayment distribution and vary the importance of persistent borrower heterogeneity and dependence in the decision-specific errors.

Data generating process. Suppressing the income state indicator for simplicity, latent repayment in the Private and Audience games is generated as

$$\begin{aligned} y_{ir}^{P*} &= \mu_i + \varepsilon_{ir}^P, \\ y_{ir}^{A*} &= \mu_i + \beta + \varepsilon_{ir}^A, \end{aligned} \tag{12}$$

where μ_i is the individual specific latent repayment level and β is the audience effect on latent repayment. Each simulated sample contains 919 individuals observed in three rounds under both experimental conditions, giving 5,514 observations. I set $\beta = 1,000$ Taka throughout the simulation so that differences across simulation cells reflect differences in the data generating process rather than changes in the underlying audience effect. Observed repayment is censored at the experimental bounds, $y_{ir}^g = \min\{\max(y_{ir}^{g*}, 0), \bar{Y}\}$, where $\bar{Y} = 35,000$ Taka and $g \in \{P, A\}$.

Both experimental conditions have total latent repayment variance $\text{Var}(y_{ir}^{P*}) = \text{Var}(y_{ir}^{A*}) = \sigma^2 = \sigma_\mu^2 + \sigma_\varepsilon^2$, and $\text{Var}(\varepsilon_{ir}^P) = \text{Var}(\varepsilon_{ir}^A) = \sigma_\varepsilon^2$.

For a given σ^2 , let ϕ_μ denote the share of total latent repayment variance attributable to individual heterogeneity.

$$\phi_\mu \equiv \frac{\sigma_\mu^2}{\sigma^2}, \quad \sigma_\mu^2 = \phi_\mu \sigma^2, \quad \sigma_\varepsilon^2 = (1 - \phi_\mu) \sigma^2.$$

Individual specific repayment levels are distributed as

$$\mu_i \stackrel{\text{iid}}{\sim} N(\bar{\mu}, \sigma_\mu^2),$$

independently of the decision-specific errors.

I allow the decision specific errors to have an exchangeable correlation structure across rounds. For each individual i , let $\varepsilon_i = (\varepsilon_{i1}^P, \varepsilon_{i1}^A, \dots, \varepsilon_{i3}^P, \varepsilon_{i3}^A)^\top$. The error vectors are independently and identically distributed across individuals according to

$$\varepsilon_i \sim N(\mathbf{0}, \sigma_\varepsilon^2 \Sigma),$$

where the (r, r') block of Σ is

$$\Sigma_{rr'} = \begin{cases} \begin{pmatrix} 1 & \rho_\varepsilon \\ \rho_\varepsilon & 1 \end{pmatrix} & \text{if } r = r', \\ \begin{pmatrix} \rho_{\text{round}} & \rho_{\text{round}} \rho_\varepsilon \\ \rho_{\text{round}} \rho_\varepsilon & \rho_{\text{round}} \end{pmatrix} & \text{if } r \neq r'. \end{cases}$$

The parameter ρ_ε determines the within-round cross-game correlation in idiosyncratic deviations, while ρ_{round} governs the cross-round correlation in idiosyncratic deviations within the same game. Thus, $\text{Cor}(\varepsilon_{ir}^g, \varepsilon_{ir'}^g) = \rho_{\text{round}}$ for $g \in \{P, A\}$ and $r \neq r'$, while $\text{Cor}(\varepsilon_{ir}^P, \varepsilon_{ir'}^A) = \rho_\varepsilon \rho_{\text{round}}$ for $r \neq r'$. Under this separable correlation structure, the cross-round cross-game correlation is the product of the two parameters. Setting $\rho_{\text{round}} = 0$ allows independence across rounds while retaining the within-round cross-game correlation ρ_ε , as

$$\begin{pmatrix} \varepsilon_{ir}^P \\ \varepsilon_{ir}^A \end{pmatrix} \stackrel{\text{iid}}{\sim} N \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \sigma_\varepsilon^2 \begin{pmatrix} 1 & \rho_\varepsilon \\ \rho_\varepsilon & 1 \end{pmatrix} \right].$$

Calibration and parameters. I calibrate $\bar{\mu}$ and σ to the pooled lower- and upper-censoring shares in the Normal state. Because the sample contains equal numbers of Private and Audience observations, the two latent distributions, $N(\bar{\mu}, \sigma^2)$ and $N(\bar{\mu} + \beta, \sigma^2)$, receive equal weight. Matching the empirical shares at zero and 35,000 gives $\bar{\mu} = 26,625$ and $\sigma = 21,668$. I then vary ϕ_μ , ρ_ε , and ρ_{round} , using 500 replications per cell. Each simulated sample contains 919 borrowers observed in three rounds under both conditions, giving 5,514 observations.

Table E.1: Monte Carlo Parameterization

Parameter	Symbol	Value	Role
Borrowers	n	919	Simulated estimation sample
Rounds	r	3	Repeated decisions per individual in each game
True latent audience effect	β	1,000	Target latent estimand
Mean latent repayment level	$\bar{\mu}$	26,625	Calibrated mean of latent repayment
Total standard deviation	σ	21,668	Calibrated dispersion of latent repayment
Individual variance share	ϕ_μ	0.30, 0.50, 0.70, 0.90	Share of latent variance from individual heterogeneity
Cross-game error correlation	ρ_ε	-0.9, -0.6, -0.3, 0.00, 0.3, 0.6, 0.9	Correlation of Private and Audience errors within a round
Cross-round error correlation	ρ_{round}	0.00, 0.3, 0.6, 0.9	Correlation across rounds
Experimental bounds	$[0, \bar{Y}]$	$[0, 35,000]$	Lower and upper repayment limits
Replications	B	500	Number of Monte Carlo replications

Estimators and estimands. I compare three estimators. First, the linear fixed effects specification is

$$y_{igr} = v_i + \beta^F \text{Audience}_{igr} + u_{igr},$$

where β^F estimates the observed audience effect β^O .

The two-limit Tobit specification is

$$y_{igr}^* = \alpha + \beta^T \text{Audience}_{igr} + u_{igr}, \quad y_{igr} = \min\{\max(y_{igr}^*, 0), \bar{Y}\},$$

where β^T estimates the latent audience effect β .

The audience effect is subject to an additional layer of censoring beyond the bounds on repayment itself. Because repayment is bounded between zero and $\bar{Y}$, the observed difference between Audience and Private repayment is also bounded, with the feasible range depending on the borrower's Private-game repayment. In particular, borrowers with high Private repayment have less room to increase observed repayment in the Audience game. This feature makes a variable-limit Tobit a natural alternative for estimating the latent audience effect, since the censoring limits of the repayment difference vary across borrowers. I therefore include this estimator in the Monte Carlo exercise to evaluate its performance, although I do not use it in the main analysis. The variable-limit Tobit defines $\tilde{y}_{ir} = y_{ir}^A - \bar{y}_i^P$, where $\bar{y}_i^P = \frac{1}{3} \sum_r y_{ir}^P$, and estimates

$$\tilde{y}_{ir}^* = \beta^V + u_{ir}, \quad \tilde{y}_{ir} = \min\{\max(\tilde{y}_{ir}^*, -\bar{y}_i^P), \bar{Y} - \bar{y}_i^P\},$$

where β^V is interpreted as the latent audience effect relative to the observed Private-game repayment benchmark.

Results. Figure E.1 reports the simulation results. The fixed effects estimator remains centered around the observed effect, $\beta^O = 537$, with deviations between -3.8 and $+1.5$ percent across the simulation grid. The two-limit Tobit similarly remains centered around the latent effect, with deviations between -3.7 and $+1.6$ percent of β . The correlation parameters alter dependence across repeated observations but leave the marginal latent distributions unchanged, and neither point estimate is materially affected by this dependence.

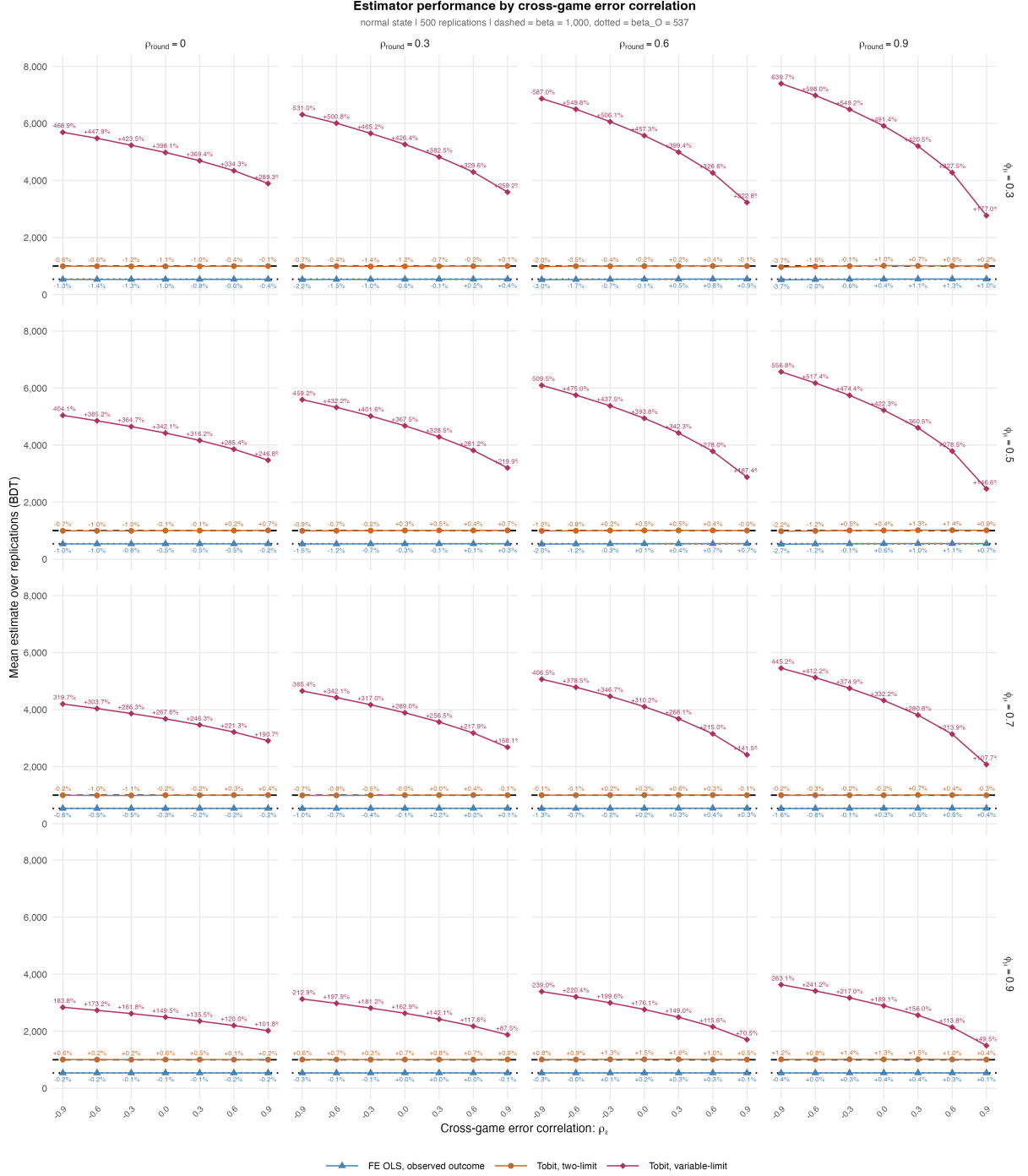
The two estimators therefore recover different quantities. Under the Normal-state calibration, $\beta^O / \beta = 0.537$, so censoring reduces the observed repayment response to a little over half of the latent response. For a small shift in latent repayment, this ratio is approximately the uncensored share of the repayment distribution, which is also 0.537 in the calibration. The fixed effects and Tobit estimates should therefore be interpreted as the audience effects on observed and latent repayment, respectively, rather than as competing estimates of the same parameter.

The variable-limit Tobit, by contrast, substantially overstates β . The observed Private benchmark $\bar{y}_i^P$ is a censored and noisy proxy for μ_i , so the latent transformed outcome is $y_{ir}^{A*} - \bar{y}_i^P = \beta + (\mu_i - \bar{y}_i^P) + \varepsilon_{ir}^A$ rather than β plus a mean-zero error. I therefore do not use this specification in the main analysis.

Overall, the simulation establishes that, under the maintained censored-normal data generating process, the fixed effects and two-limit Tobit estimators recover their intended observed and latent

audience effects across a wide range of persistent borrower heterogeneity and within-borrower error dependence.

Figure E.1: Monte Carlo Performance of Censored Outcome Estimators



Notes. The figure reports mean estimates across 500 Monte Carlo replications under the Normal-state calibration. The dashed line denotes the true latent audience effect, $\beta = 1,000$, and the dotted line denotes the implied observed audience effect, $\beta^O = 537$. Rows vary the individual variance share ϕ_μ , columns vary the cross-round error correlation ρ_{round} , and the horizontal axis varies the within-round cross-game error correlation ρ_ϵ . Percentage labels report bias relative to the corresponding target.